

# From Multi-Agent to Single-Agent: When Is Skill Distillation Beneficial?

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Project Page <https://adaptive-skill.github.io/>  
Code <https://github.com/Tencent/AdaSkill>

## Abstract

Multi-agent systems (MAS) tackle complex tasks by distributing expertise, though this often comes at the cost of heavy coordination overhead, context fragmentation, and brittle phase ordering. Distilling a MAS into a single-agent skill can bypass these costs, but this conversion lacks a principled answer for *when* and *what* to distill. Instead, the empirical outcome is surprisingly inconsistent: skill lift ranges from a 28% improvement to a 2% degradation across metrics of the exact same task. In this work, we reveal that skill utility is governed not by the task, but by the evaluation metric. We introduce *Metric Freedom* ( $F$ ), the first *a priori* predictor of skill utility.  $F$  measures the topological rigidity of a metric’s scoring landscape by quantifying how output diversity couples with score variance via a Mantel test. Guided by  $F$ , we propose **AdaSkill**, a two-stage adaptive distillation framework. Stage 1 acts as a selective extraction mechanism, extracting tools and knowledge while discarding restrictive structures on “free” metrics to preserve exploration. Stage 2 applies iterative refinement selectively on free metrics, exploiting their forgiving scoring landscape to safely maximize remaining headroom. Evaluating across 4 tasks, 11 datasets, and 6 metrics,  $F$  strongly predicts skill utility ( $r=-0.85$ ,  $p<0.0001$ ). Strikingly, identical agent trajectories yield diametrically opposite skill lifts under rigid versus free metrics, demonstrating that skill utility is fundamentally a metric-level property. Driven by this signal, AdaSkill matches or exceeds the original MAS while reducing cost up to  $8\times$  and latency by up to  $15\times$ .

## 1 Introduction

Large Language Model (LLM) agents have demonstrated remarkable proficiency across complex domains, from scientific discovery to software engineering (Boiko et al., 2023; Jimenez et al., 2024; Xu et al., 2026c; Zhou et al., 2024). To tackle increasingly difficult tasks, the field has largely converged on Multi-Agent Systems (MAS), which decompose problems into specialized, interacting sub-agents to capture domain expertise. However, this distributed approach comes at a steep cost: severe coordination overhead, context fragmentation (Xu et al., 2026b), and brittle phase ordering. Consequently, a compelling alternative has emerged: distilling the expertise of a MAS into a single-agent system equipped with specialized skills (Luo et al., 2026; Xu et al., 2026f). While recent work shows these single-agent systems can match MAS performance at a fraction of the cost (Xu et al., 2026a), empirical results reveal a baffling inconsistency. Skill distillation can yield massive performance gains (+28pp) or actively degrade performance (−2pp) on the *exact same task*.

Existing literature largely treats this variance as an unpredictable byproduct of task complexity or data scale, proceeding under the assumption that skill augmentation is universally beneficial. Current approaches focus entirely on *how* to distill skills: either through rigid structural conversions that map MAS pipelines into instructions (Li, 2026; Ni et al., 2026; Yim et al., 2026), or through iterative trajectory optimization (Alzubi et al., 2026; Xia et al., 2026; Yang et al., 2026; Zhou et al., 2026).

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Both paradigms suffer from a critical blind spot. Non-adaptive structural conversions routinely over-constrain agents on simple queries, while iterative methods succumb to trajectory-local overfitting, bloating the skill library until retrieval accuracy collapses (Li, 2026). The fundamental question remains unanswered: *when does distilling a MAS into single-agent skills actually help?*

In this work, we reveal that the efficacy of skill distillation is not dictated by the task itself, but by an overlooked property of the *evaluation metric*. We introduce **Metric Freedom** ( $F$ ), the first quantifiable, *a priori* predictor of skill utility.  $F$  measures the topological rigidity of a metric’s scoring landscape by quantifying how tightly output variation couples with score variation via a Mantel test. On **rigid** metrics (low  $F$ ), structured skill guidance steers the agent through a narrow success corridor and yields large gains. On **free** metrics (high  $F$ ), diverse paths score equally well, so the same constraints restrict exploration and degrade performance.

To achieve this insight, we propose **AdaSkill**, an  $F$ -guided two-stage adaptive distillation framework. Rather than blindly mimicking MAS pipelines, our approach uses Metric Freedom as a principled sieve to determine exactly *what* to extract and *when* to refine it. In **Stage 1 (Adaptive MAS Converter)**, tools are universally preserved and domain knowledge is distilled into conditional references rather than rigid, mandatory instructions. Coordination structures are discarded entirely, and explicit task-decomposition hints are retained *only* for low- $F$  metrics. **Stage 2 (Skill Iterator)** introduces an optional four-agent iterative loop (Explore, Main, Analyzer, Runner) to close the residual gap left by Stage 1’s conservative extraction. This loop is activated only on mid- and high- $F$  metrics, where the forgiving landscape permits safe, monotonic improvement. On low- $F$  metrics, the knife-edge landscape turns iterative fixes into zero-sum oscillations, so Stage 2 is explicitly disabled.

Extensive evaluation across four complex tasks, 11 datasets, and 6 diverse metrics validates our core hypothesis: Metric Freedom strongly and reliably predicts the utility of skill distillation ( $r = -0.85$ ,  $p < 0.0001$ ,  $n = 18$ ,  $F_{\text{out}}$ ). Notably, the MAS Compiler, which faithfully converts MAS pipelines without  $F$ -guidance, frequently underperforms even the unskilled Base Agent. By acting on the  $F$  signal alone, AdaSkill matches or exceeds the original MAS while reducing inference cost by up to  $8\times$  and latency by up to  $15\times$ . Our core contributions are:

- **Metric Freedom ( $F$ ):** We introduce the first *a priori*, metric-level criterion for predicting when MAS-to-skill conversion is beneficial.  $F$  is computable entirely from standard baseline runs, requiring no additional skilled-agent inference overhead.
- **AdaSkill (Adaptive Two-Stage Distillation):** We design a principled framework that abandons the “one-size-fits-all” approach to skill generation. Guided by  $F$ , Stage 1 selectively distills knowledge and structures. Stage 2 targets refinement toward mid- and high- $F$  metrics where the forgiving landscape permits stable, monotonic improvement, and explicitly disables iteration on low- $F$  metrics where the knife-edge landscape turns fixes into zero-sum oscillations.
- **Reframing Skill Evaluation:** We provide conclusive evidence that skill utility is a *metric-level* property, not a task-level one. By demonstrating that identical agent trajectories can yield diametrically opposite skill lifts under rigid versus free metrics, we fundamentally correct a prevalent blind spot in how agentic skills are evaluated.

## 2 Background and Related Work

**Multi-Agent Systems.** The rise of tool-augmented agents (Schick et al., 2023; Yao et al., 2023) and frameworks like AutoGen (Wu et al., 2024) and MetaGPT (Hong et al., 2024) has spurred task-specific Multi-Agent Systems (MAS). Recent MAS excel in specialized domains, such as text-to-SQL via hypothesis-verification loops (APEX-SQL (Cao et al., 2026); evaluated on BIRD (Li et al., 2023)), causal discovery combining agent debate with classical algorithms (Spirtes et al., 2012; Zheng et al., 2018) (MATMCD (Shen et al., 2025)), causal estimation automation (Imbens and Rubin, 2015) (CAIS (Verma et al., 2025)), and feature engineering (FELA (Ouyang et al., 2025)). However, these systems rely on rigid phase ordering, inter-agent messaging, and specialized sub-agents. This structural complexity introduces overhead that may be unnecessary given that capable single models can perform internal multi-perspective reasoning (Qian et al., 2024).

**Single-Agent vs. Multi-Agent.** Recent studies challenge the assumption that MAS consistently outperform single agents. Li (2026) show single agents with skill libraries can match MAS cost-effectively, though unstructured accumulation causes a performance-degrading phase transition. Xu

et al. (2026f) shows that single-agent multi-turn execution (OneFlow) rivals MAS across diverse tasks. Meanwhile, Luo et al. (2026) extends classical knowledge distillation (Hinton et al., 2015) to compress multi-agent interactions into a single model (AgentArk), and Aluru et al. (2025) distills multi-agent debate interaction graphs into a single student model via graph-aware training. Our work differs from all these: rather than comparing paradigms broadly or distilling interaction patterns, we investigate *which specific MAS components* warrant conversion and provide a metric-level criterion ( $F$ ) that predicts when such conversion is beneficial. Addressing the skill-quantity bottleneck (Li, 2026), our framework constructs task-adapted,  $F$ -guided skills instead of indiscriminately accumulating them.

**Skill Discovery and Evolution.** Skill-level optimization is crucial for transferable capabilities, as demonstrated by Alzubi et al. (2026) via EvoSkill’s iterative failure analysis and Voyager (Wang et al., 2023) via open-ended embodied exploration. Similarly, Ni et al. (2026) hierarchically consolidate agent experiences into declarative skills (Trace2Skill), overcoming manual authoring bottlenecks. In multi-agent RL, Zhou et al. (2025) use dual distillation modules to compress agent interactions into a shared policy network, and Wang et al. (2026) propose skill-conditioned self-distillation for multi-turn LLM agents. While our skill iterator (Stage 2) shares the spirit of iterative refinement, it uniquely operates on *MAS-derived* skills rather than discovering them from scratch, and uses Metric Freedom to decide *what* and *how much* to distill rather than applying a fixed distillation strategy.

### 3 Methodology

#### 3.1 Overview

The core premise of our framework is that distilling MAS on a skill is a double-edged sword: it guides reasoning, but it inherently constrains exploration. To balance this, we propose that *such structural guidance should only be imposed where the evaluation metric prefers “a single standard answer”*. To operationalize this, we introduce **Metric Freedom** ( $F$ ), a formal measure of a metric’s scoring landscape. This concept drives our two-stage system: an *adaptive MAS converter* (Stage 1) that selectively extracts structure only where  $F$  dictates it is beneficial, and a *skill iterator* (Stage 2) that performs targeted, automated refinement strictly in higher- $F$  regimes where the scoring landscape permits safe exploration.

#### 3.2 Metric Freedom: Formalizing the Scoring Landscape

Not all evaluation metrics benefit equally from structured guidance. In causal estimation, Method Selection Accuracy (MSA) is all-or-nothing, whereas Mean Relative Error (MRE) rewards proximity to the ground truth. We formalize this via Metric Freedom ( $F$ ), a quantity measuring the topological decoupling between an agent’s behavioral variation and its score variation.

**Definition & Formulation.** Given  $n$  independent baseline runs, let  $\mathcal{X}$  be a behavioral representation space equipped with a pairwise distance function  $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, 1]$ . Each run  $i$  yields a triple  $(x_i^{\text{out}}, x_i^{\text{trace}}, s_i)$ : a final output, a reasoning trace, and a score. We define a shared score distance matrix  $D_{ij}^{\text{score}} = |s_i - s_j|$ , and two behavioral distance matrices  $D_{ij}^{\text{out}} = d(x_i^{\text{out}}, x_j^{\text{out}})$  and  $D_{ij}^{\text{trace}} = d(x_i^{\text{trace}}, x_j^{\text{trace}})$ . To robustly capture the topological relationship without distributional assumptions, we apply the Mantel test (Mantel, 1967). Let  $\mathcal{X}$  denote either behavioral space, and let  $D^{\mathcal{X}}$  denote the corresponding distance matrix. Let  $R_{ij}^{\mathcal{X}} = \text{rank}(D_{ij}^{\mathcal{X}})$  and  $R_{ij}^{\text{score}} = \text{rank}(D_{ij}^{\text{score}})$  denote the ranks of the upper-triangular elements. We compute Spearman’s rank correlation  $r_M \in [-1, 1]$  as:

$$r_M(\mathcal{X}) = \frac{\sum_{i < j} (R_{ij}^{\mathcal{X}} - \bar{R}^{\mathcal{X}})(R_{ij}^{\text{score}} - \bar{R}^{\text{score}})}{\sqrt{\sum_{i < j} (R_{ij}^{\mathcal{X}} - \bar{R}^{\mathcal{X}})^2 \cdot \sum_{i < j} (R_{ij}^{\text{score}} - \bar{R}^{\text{score}})^2}}. \quad (1)$$

Metric Freedom is then defined as the orthogonal complement to this concordance:

$$F_{\mathcal{X}} = 1 - r_M(\mathcal{X}). \quad (2)$$

Intuitively,  $F_{\mathcal{X}} \approx 0$  indicates a “knife-edge” scoring surface: behavioral variation perfectly predicts score variation, meaning there is a narrow, rigid corridor of successful behavior. Conversely, high

$F_{\mathcal{X}}$  implies a flat scoring surface: the agent can explore broadly without score penalty. Note that  $F \in [0, 2]$  since  $r_M \in [-1, 1]$ ; In practice,  $F > 1$  represents extreme high freedom.

**Instantiations & Empirical Reproducibility.** We compute  $F$  from  $n$  base agent runs over  $M$  questions sampled uniformly from the benchmark. To mitigate output collapse during standard inference, each run is seeded with a methodologically distinct prior generated by a lightweight *diversity planner*. This ensures behavioral diversity without introducing bias. Sensitivity analysis in Appendix B.2 confirms that this approach primarily enhances sampling efficiency: while metric ordering remains unbiased compared to independent random runs, the planner significantly reduces variance at low  $n$  (e.g.,  $< 5\%$  vs.  $20\%$  error at  $n = 6$ ). Finally, we extract only *mixed questions* (excluding 100% success/failure cases to isolate from task difficulty):

- **Output freedom** ( $F_{\text{out}}$ ):  $\mathcal{X}$  is the space of final outputs, and  $d$  is a task-appropriate structural distance (e.g., Jaccard for sets, Hamming for categories; see Table 1).
- **Trace freedom** ( $F_{\text{trace}}$ ):  $\mathcal{X}$  is the semantic embedding space of the full reasoning trace (intermediate thoughts, tool calls) embedded via `text-embedding-3-large` (Achiam et al., 2023). Distance is  $d(\tau_i, \tau_j) = 1 - \text{cos}_{\text{sim}}(\text{embed}(\tau_i), \text{embed}(\tau_j))$ .

These measures correlate strongly ( $r \approx 0.78$ ; Appendix B.3). Confidence intervals are established via bootstrap resampling ( $B=1000$ ) over run pairs, ensuring robustness across different distance function choices. To isolate the true efficacy of structured guidance from baseline performance ceilings, we report *headroom-normalized lift*:  $\text{lift}_{\text{norm}} = \text{lift} / (1 - \text{baseline})$ .

### 3.3 Theoretical Foundation

We show that Metric Freedom controls an upper bound on the gain achievable by any skill. Let  $s : \mathcal{X} \rightarrow [0, 1]$  be a Hölder continuous scoring function,  $\mathbb{P}_0$  and  $\mathbb{P}_\pi$  the output distributions of the base and skill-augmented agents, and  $\text{Lift}(\pi) \triangleq \mathbb{E}_{\mathbb{P}_\pi}[s] - \mathbb{E}_{\mathbb{P}_0}[s]$ .

**Theorem 3.1** (Bound on Skill Lift). *Assume  $s$  is  $(\alpha, L_0)$ -Hölder continuous on  $(\mathcal{X}, d)$  with  $\alpha \in (0, 1]$ , and that the metric exhibits non-negative concordance ( $r_M \geq 0$ ). Then for skill  $\pi$ :*

$$\text{Lift}(\pi) \leq L_0 (1 - F + \Delta_n) \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_\pi, \mathbb{P}_0), \quad (3)$$

where  $\tilde{\mathcal{W}}_1$  is the Wasserstein-1 distance under the lifted metric  $\tilde{d}(x, y) = d(x, y)^\alpha$ , and  $\Delta_n \leq 0.047 \rho_m + O(n^{-1/2})$  is a small finite-sample correction ( $\rho_m$  is the HGR maximal correlation; see Proposition A.5 for the explicit characterization of  $\Delta_n$  and the origin of the 0.047 constant; formal proof in Appendix A.1). No absolute continuity ( $\mathbb{P}_\pi \ll \mathbb{P}_0$ ) is required. The Hölder condition is likewise mild: for discrete metrics such as MSA,  $L_0 = 1$  follows automatically (Appendix A.1).

*Proof intuition.* We decompose  $s$  into a component  $s_{\parallel}$  concordant with the behavioral metric and a discordant residual  $s_{\perp}$  via  $L^2$ -projection onto the Lipschitz subspace. The concordant lift is bounded by  $L_0 \rho_m \cdot \tilde{\mathcal{W}}_1$  using the HGR variational characterization, and the discordant residual contributes at most  $L_0(1 + \rho_m) \cdot \tilde{\mathcal{W}}_1$  by the triangle inequality for Lipschitz constants. The classical inequality  $|r_M| \leq \rho_m$  then links the combined bound to  $F = 1 - r_M$ , yielding a gain ceiling of  $L_0(1 - F)$  up to a universal constant. Full proof in Appendix A.1.  $\square$

**Design Implications.** The gain ceiling vanishes as  $F \rightarrow 1$ : skill distillation is fundamentally limited on free metrics. When  $F > 1$ , the bound reduces to the trivial Lipschitz constant (Remark A.1), yet the empirical trend extends smoothly across the entire  $F$  range ( $r = -0.85$ ,  $p < 0.0001$ ; Section 4.3). We further show the bound is tight (Appendix A.2), establish a phase transition explaining why iterative refinement oscillates in low- $F$  but converges in high- $F$  regimes (Appendix A.3), and prove formal properties of  $F$  including range, monotonicity, stability, and a product rule for multi-metric tasks (Appendix A.4).

### 3.4 Stage 1: Adaptive MAS Converter

Guided by Theorem 3.1, Stage 1 is a compiler that transforms a MAS into a single-agent skill. Rather than blindly copying the MAS pipeline, it extracts components based on  $F$ .

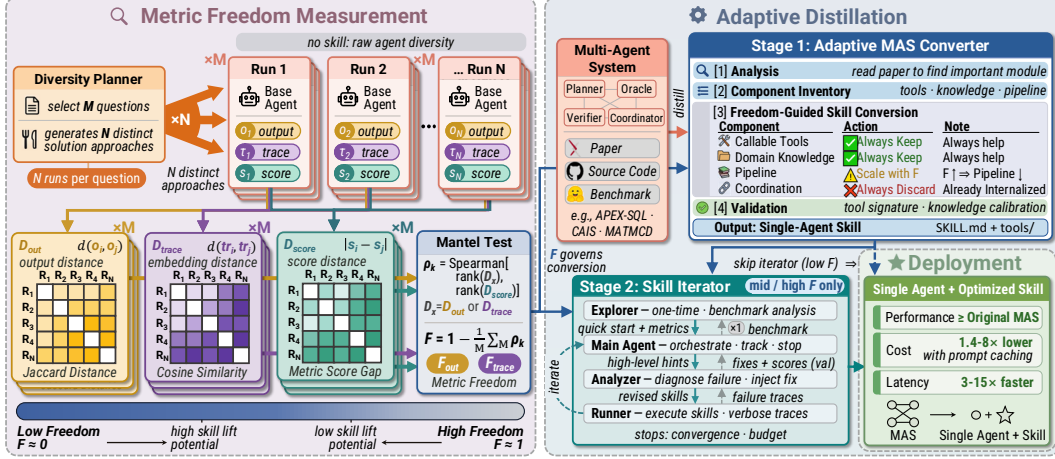


Figure 1: AdaSkill system overview.

**Table 1:** Output distance functions per task used in the Mantel test for  $F_{out}$  computation. For  $F_{trace}$ , reasoning trace scores are embedded via cosine distance replaces  $d(\cdot, \cdot)$  uniformly across all tasks.

| Task                | Output                    | $d(o_i, o_j)$ for $F_{out}$                        | Aggregation                        |
|---------------------|---------------------------|----------------------------------------------------|------------------------------------|
| Causal Discovery    | Edge set $E$              | $1 - \text{Jaccard}(E_i, E_j)$                     | Per-dataset, $F$ over all $N$ runs |
| Causal Estimation   | Selected method           | $\mathbf{1}[m_i \neq m_j]$                         | Per-question, median $F$           |
| Feature Engineering | Feature set $\mathcal{F}$ | $1 - \text{Jaccard}(\mathcal{F}_i, \mathcal{F}_j)$ | Per-dataset, $F$ over all $N$ runs |
| Text-to-SQL         | Predicted SQL             | $1 - \text{Jaccard}(\text{tok}_i, \text{tok}_j)$   | Per-question, median $F$           |

**Phases 1 & 2: Analysis and Inventory.** The distiller decomposes the MAS architecture into four component classes: (a) callable tools, (b) domain knowledge, (c) pipeline structure, and (d) coordination mechanisms.

**Phase 3:  $F$ -Guided Conversion.** Two rules are metric-invariant: coordination mechanisms are *always discarded* (a single large model internalizes coordination natively), and callable tools are *always retained* (they extend capability without constraining exploration). The treatment of **pipeline structure** follows  $F$  monotonically: as  $F$  increases the effective lift ceiling  $L_0(1-F)\hat{W}_1$  decreases (Theorem 3.1), so the distiller scales structural rigidity accordingly—from mandatory step sequences for near-zero- $F$  metrics, through conditional hints at intermediate values, to bare-minimum reference knowledge when  $F$  approaches or exceeds 1. The ablation in Section 4.5 empirically validates this monotone relationship: pipeline structure alone yields  $r=-0.88$  with lift across our metrics (Figure 6), confirming that structural imposition is the primary driver of harm at high  $F$ . For multi-metric tasks, distillation is applied per metric: structure is imposed in proportion to each metric’s  $F$ , and the resulting modules are composed into a single skill.

**Phase 4: Validation.** The distiller automatically verifies tool imports, ensures no sequences leak into high- $F$  skills, validates neutral reference tones, and tests boundary conditions (Fig. 1, Panel C).

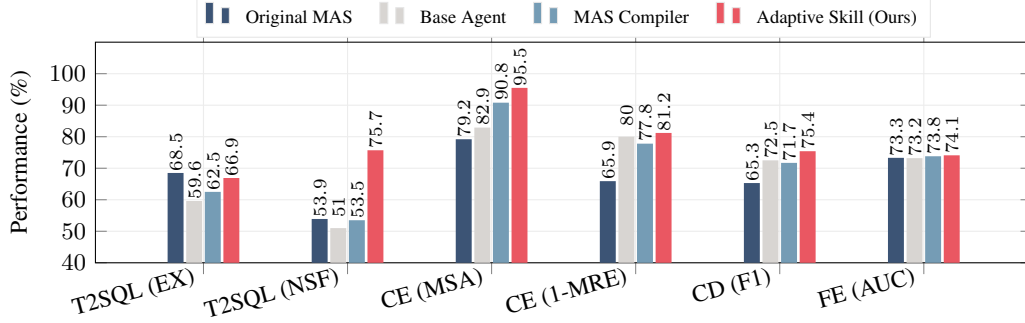
### 3.5 Stage 2: Skill Iterator

Stage 1 intentionally leaves a gap below the lift ceiling at higher  $F$  tiers by conservatively stripping structure. Stage 2 is an optional refinement module to close this gap.

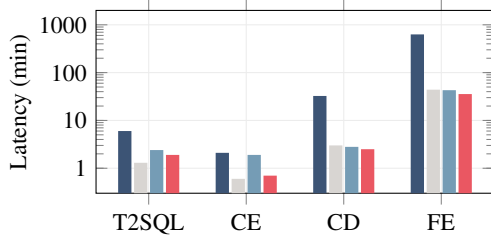
Crucially, **Stage 2 is only activated for Mid and High- $F$  metrics**. In Low- $F$  environments, the sharp landscape causes empirical *oscillation* (Section 4.6). In higher- $F$  regimes, the forgiving landscape permits safe, monotonic hill-climbing.

**Four-Agent Architecture.** The autonomous loop employs four roles (Figure 1, Panel D):

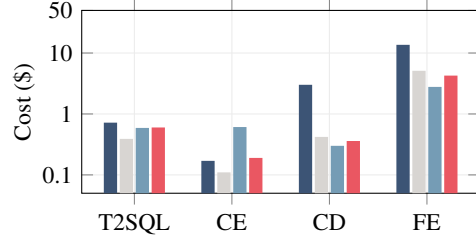
- **Explore Agent** runs once to analyze benchmark boundaries and dataset characteristics.
- **Main Agent** orchestrates runs, tracks evolution history, and makes stopping decisions.



(a) Task-averaged performance.



(b) Task-averaged latency in minutes.



(c) Task-averaged cache cost.

**Figure 2: Task-level performance, cost, and latency overview.** AdaSkill matches or outperforms all baselines on accuracy while preserving cost and latency. Per-dataset breakdowns in Appendix D.

- **Runner** executes the skill on train/val splits, generating verbose reasoning traces.
- **Analyzer** is spawned *statelessly* per iteration to diagnose the 1–3 most severe failure traces. Zero memory between iterations prevents compounding hallucinations. It classifies root causes and atomically injects fixes into tool coverage or thinking pathways.

**Termination.** The loop terminates upon convergence, sufficiency ( $> 95\%$  accuracy), oscillation, or budget exhaustion, outputting the Pareto-optimal skill version for deployment.

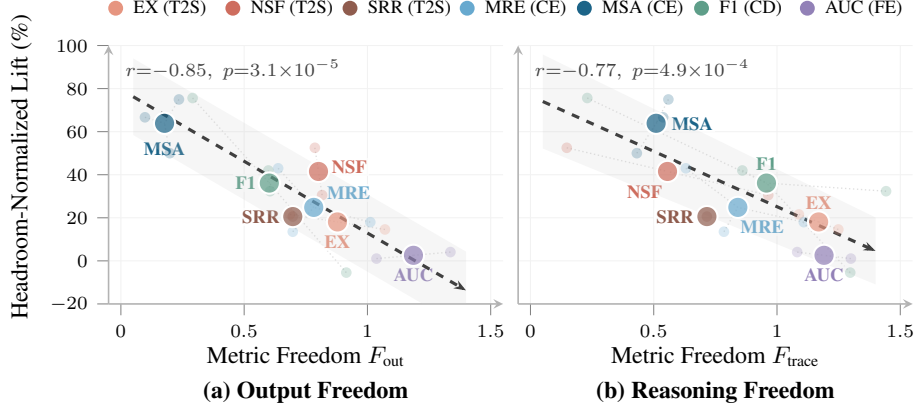
## 4 Experiments

### 4.1 Setup

**Tasks and datasets.** We evaluate on 4 tasks with 11 datasets: **Text-to-SQL** on BIRD-147 (Li et al., 2023) and Spider-120 (Lei et al., 2025); **Causal Estimation (CE)** on QRData, Synthetic, and Real datasets from CauSciBench (Acharya et al., 2026); **Causal Discovery (CD)** on AutoMPG (Quinlan, 1993), DWDClimat (Mooij et al., 2016), Sachs (Sachs et al., 2005) and Child (Spiegelhalter, 1992); and **Feature Engineering (FE)** on Taobao (Alibaba Group, 2018) and Dia (Teboul, 2022).

**Methods.** We compare 5 paradigms: **Original MAS** (APEX-SQL (Cao et al., 2026), CAIS (Verma et al., 2025), MATMCD (Shen et al., 2025), FELA (Ouyang et al., 2025)—representing SOTA multi-agent frameworks), **Base Agent** (raw single agent, no skill), **Adaptive Skill (Ours)**, **Auto Optimization** (Alzubi et al., 2026), and **MAS Compiler** (a baseline that compiles all MAS components faithfully) (Li, 2026). MAS baselines follow official implementation. To ensure fairness, all methods use Claude Sonnet 4.6 as the backbone model, and all single-agent methods use Claude Code backend.

**Evaluation protocol.**  $F$  is computed from  $n=6$  raw-agent runs (without skill) per question and  $m = 6$  questions per dataset. For per-question tasks (CE, T2SQL),  $F$  is the median over *mixed questions*. Stage 1 uses full code, paper, and benchmark from each task to build adaptive skill. Stage 2 uses a held-out 10% for training and validation each to track per-iteration gains.



**Figure 3: Metric freedom  $F$  predicts performance lift of skills.** Both output-space (a) and reasoning-space (b) measures confirm the negative trend ( $r = -0.85$  and  $r = -0.77$ ), validating  $F$  as a predictor of skill utility. Large circles = metric-level aggregates; small circles = individual datasets.

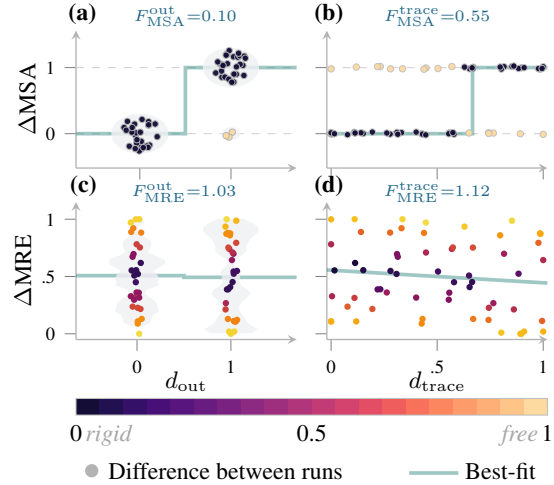
## 4.2 Main Results: Performance, Latency, and Cost

Fig. 2 provides a unified view of performance, cost, and latency. Across all four domains, our AdaSkill consistently reaches a balanced performance. It matches or exceeds the performance of Original MAS baselines while operating at a fraction of the cost (up to  $8\times$  cheaper) and latency ( $3\text{--}15\times$  faster). Crucially, the MAS Compiler blindly translates MAS pipelines into single-agent skills yet often underperforms both the Base Agent and our Adaskill, proving that simply copying agentic structure is suboptimal without  $F$ -guided extraction. Beyond accuracy, prompt caching further widens the efficiency gap: single-agent methods cost 30–40% of their uncached price, whereas Original MAS baselines receive near-zero cache benefit due to dynamic inter-agent messaging. Per-dataset efficiency breakdowns across all 11 datasets are in Appendix D.

## 4.3 Metric Freedom as a Predictor

**Global Correlation.** Fig. 3 plots headroom-normalized lift against  $F$  across all tasks and metrics. Critically,  $F$  is computed entirely from raw-agent runs before any skill is applied—making it a genuine a priori predictor of the metric’s intrinsic ceiling on path-controllable improvement. Both output-space and trace-space freedom confirm a strong negative trend:  $r(F_{\text{out}}, \text{lift}_{\text{norm}}) = -0.85$  ( $p < 0.0001$ ) and  $r(F_{\text{trace}}, \text{lift}_{\text{norm}}) = -0.77$  ( $p < 0.001$ ), consistent across all 4 domains and 6 metrics spanning rigid scores ( $F \approx 0$ ) to diffuse landscapes ( $F \approx 0.9$ ). In practice, this means skill utility on a new task can be predicted from raw runs alone—without deploying a single skilled agent; Results on GPT-5.1 replication yield consistent trends ( $r = -0.71$ ,  $p < 0.01$ ; Appendix B.6).

**$F$  Computation: A Within-Task Illustration.** Fig. 4 illustrates the  $F$  computation on the same 60 raw-agent run-pairs from Causal Estimation, evaluated under two metrics. Each point plots path distance between two runs ( $x$ ) against their metric difference ( $y$ ); a tight fit indicates that path variation reliably predicts



**Figure 4:  $F$  quantifies whether path differences predict metric outcomes.** Each point pairs two raw agent runs:  $x$  = path distance,  $y$  = metric difference. A tighter fit yields lower  $F$ , indicating that skill-based path control reliably shifts the metric.



**Table 2: Causal Estimation results.** The same agent outputs yield opposite lift directions under MSA ( $F \approx 0.2$ ) and MRE ( $F \approx 0.8$ ), providing evidence that skill utility is a metric-level property.

| Dataset→<br>Method↓      | Textbook    |             | Synthetic  |            | Real        |             | Efficiency |            |
|--------------------------|-------------|-------------|------------|------------|-------------|-------------|------------|------------|
|                          | MSA↑        | MRE↓        | MSA↑       | MRE↓       | MSA↑        | MRE↓        | Time (s)↓  | Cost (\$)↓ |
| CAIS (MAS)               | 83.3        | 54.0        | 75.9       | 16.2       | 78.3        | 32.0        | 123        | 0.172      |
| Base Agent               | 61.5        | 31.2        | <b>100</b> | 10.6       | 87.1        | 18.3        | 34         | 0.107      |
| EvoSkill                 | 71.8        | 25.4        | <b>100</b> | 7.1        | 93.5        | 21.6        | 66         | 0.270      |
| MAS Compiler             | 82.1        | 26.3        | <b>100</b> | 12.9       | 90.3        | 17.3        | 115        | 0.608      |
| <i>Tools Only</i>        | 69.2        | 28.3        | 95.6       | 5.5        | 87.1        | 17.0        | 49         | 0.208      |
| <i>Knowledge Only</i>    | 79.5        | 25.6        | 91.1       | 6.0        | 90.3        | 18.0        | 55         | 0.305      |
| <i>Pipeline Only</i>     | 84.6        | 26.5        | <b>100</b> | 5.5        | <b>96.8</b> | 17.8        | 56         | 0.295      |
| AdaSkill w/o Opt. (Ours) | <b>89.7</b> | 29.5        | <b>100</b> | 6.3        | <b>96.8</b> | 16.7        | 43         | 0.193      |
| <b>AdaSkill (Ours)</b>   | <b>89.7</b> | <b>23.1</b> | <b>100</b> | <b>4.3</b> | <b>96.8</b> | <b>15.9</b> | 52         | 0.207      |

**Table 3: Text-to-SQL results.** AdaSkill achieves the best single-agent EX on both benchmarks. Side-by-side comparison highlights the schema-linking gains (NSF) across different complexities.

| Dataset→<br>Method↓    | BIRD-147    |             |             |             |             | Spider-120  |             |             |             |             | Efficiency |       |
|------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|------------|-------|
|                        | SRR↑        | NSR↑        | NSP↑        | NSF↑        | EX↑         | SRR↑        | NSR↑        | NSP↑        | NSF↑        | EX↑         | Time↓      | Cost↓ |
| APEX (MAS)             | <b>93.9</b> | <b>98.3</b> | 47.0        | 60.8        | <b>69.4</b> | <b>88.3</b> | <b>97.1</b> | 36.5        | 47.0        | <b>66.7</b> | 361        | 0.72  |
| Base Agent             | 71.8        | 91.8        | 90.0        | 90.5        | 59.2        | 0.8         | 8.4         | 21.7        | 11.4        | 60.0        | 75         | 0.39  |
| EvoSkill               | 76.2        | 92.5        | 94.7        | 93.0        | 66.0        | 2.5         | 15.2        | 32.5        | 20.8        | 62.5        | 97         | 0.48  |
| MAS Compiler           | 79.6        | 93.6        | 94.8        | 93.7        | 67.4        | 1.7         | 10.3        | 22.6        | 13.4        | 57.5        | 143        | 0.59  |
| <i>Tools Only</i>      | 75.5        | 92.5        | 94.7        | 92.9        | 66.0        | 3.3         | 20.5        | 45.0        | 28.5        | 63.3        | 104        | 0.37  |
| <i>Knowledge Only</i>  | 74.2        | 92.5        | 94.9        | 93.1        | 66.7        | 5.0         | 36.4        | 68.3        | 45.3        | 64.2        | 93         | 0.51  |
| <i>Pipeline Only</i>   | 68.7        | 90.7        | 94.2        | 91.7        | 64.6        | 1.7         | 12.0        | 28.3        | 16.2        | 60.8        | 124        | 0.88  |
| AdaSkill w/o Opt.      | 77.6        | 92.7        | <b>95.3</b> | 93.4        | 68.0        | 5.9         | 47.7        | 85.0        | 57.9        | 65.8        | 116        | 0.64  |
| <b>AdaSkill (Ours)</b> | 80.8        | 94.2        | 94.9        | <b>94.1</b> | 67.4        | 7.5         | 51.3        | <b>85.8</b> | <b>60.5</b> | <b>66.7</b> | 101        | 0.45  |

metric outcomes, yielding low  $F$ . Under MSA ( $F=0.10$ , panels a–b), the fit is near-perfect: the agent naturally explores diverse causal methods (IV, DiD, RDD, OLS), and since only one method is correct, runs far apart in behavior are also far apart in score. Under MRE ( $F=1.03$ , panels c–d), the same pairs scatter diffusely: multiple methods converge to numerically similar effect sizes, so behavioral distance no longer predicts score distance. The same 60 run-pairs yield  $F \approx 0$  under MSA and  $F \approx 1$  under MRE—confirming that  $F$  captures a property of the metric topology, not the task or the agent.

#### 4.4 Per-Task Results

**Causal Estimation** ( $F_{\text{MSA}} = 0.1\text{--}0.2$ ;  $F_{\text{MRE}} \approx 0.6\text{--}1.0$ ). This task provides the sharpest test (Table 3). On MSA ( $F \approx 0.2$ ), the MAS compiler imposes strict structure, achieving over +28.2 pp gain on QRData. On MRE ( $F \approx 0.8$ ), the exact same skill shows no advantage or even decrease on Synthetic. However, our AdaSkill consistently gets the best MSA and MRE on all the datasets. The MAS Compiler costs  $3\times$  more than the adaptive skill yet achieves lower performance.

**Text-to-SQL.** At midpoint ( $F=0.50$ ), our framework predicts moderate but reliable benefit (Table 3): the metric landscape is neither rigid enough for structure to be strictly necessary nor free enough to render it harmful. Consistent with this prediction, the adaptive skill provides the best Execution Accuracy among single-agent methods on both BIRD-147 and Spider-120 (+8.8pp and +5.8pp over Raw, respectively), without the cost overhead of the original MAS.

**Causal Discovery.** On Sachs ( $F=0.24$ , lowest freedom), AdaSkill achieves  $F1 = 0.95$  (+8pp over raw), the largest Stage 1 gain in this task (Table 4). On AutoMPG ( $F=0.41$ ), the Stage 1 skill slightly trails the base agent (0.63 vs. 0.67), consistent with the prediction that the base agent can already work in moderate-freedom spaces; Stage 2 recovers modestly to 0.65. On higher- $F$  DWDClimat and Child, Stage 2 yields the strongest gains (0.70 and 0.95 F1). Original MAS costs up to  $8\times$  more per run with no consistent accuracy advantage. Extended precision/SHD metrics are in Appendix C.1.



**Table 4: Causal Discovery F1 results (mean  $\pm$  std, 3 runs).** Gains track  $F$ : largest on low- $F$  Sachs; Stage 2 recovers further ground on mid/high- $F$  DWDClim and Child. Time/Cost are per-run means. Full precision/SHD metrics in Appendix C.1.

| Method                 | AutoMPG              | DWDClim.             | Sachs                | Child                | Time (s) | Cost (\$) |
|------------------------|----------------------|----------------------|----------------------|----------------------|----------|-----------|
| MATMCD (MAS)           | .74 $\pm$ .02        | .66 $\pm$ .06        | .56 $\pm$ .11        | .65 $\pm$ .05        | 1852     | 2.79      |
| Base Agent             | .67 $\pm$ .05        | .56 $\pm$ .01        | .87 $\pm$ .05        | .80 $\pm$ .17        | 186      | 2.22      |
| MAS Compiler           | <b>.70</b> $\pm$ .03 | .64 $\pm$ .08        | .67 $\pm$ .26        | .86 $\pm$ .05        | 149      | 1.46      |
| AdaSkill w/o Opt.      | .63 $\pm$ .02        | .59 $\pm$ .07        | <b>.95</b> $\pm$ .03 | .84 $\pm$ .05        | 150      | 1.89      |
| <b>AdaSkill (Ours)</b> | .65 $\pm$ .05        | <b>.70</b> $\pm$ .05 | .92 $\pm$ .06        | <b>.95</b> $\pm$ .07 | 188      | 2.50      |

**Table 5: Feature Engineering results (AUC, mean  $\pm$  std).** Under high  $F$ , all methods reach near-identical accuracy; AdaSkill matches the full FELA skill on Taobao while halving cost, confirming that minimal structure is optimal when the scoring landscape is flat.

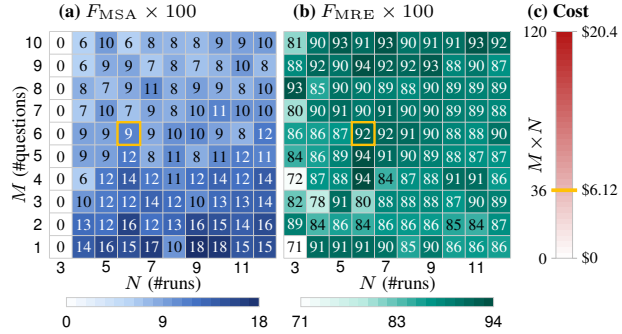
| Method                       | Taobao                 |             |    |          |                 | Dia                    |             |    |          |                 |
|------------------------------|------------------------|-------------|----|----------|-----------------|------------------------|-------------|----|----------|-----------------|
|                              | AUC                    | Max         | #F | Time (s) | Cost (\$)       | AUC                    | Max         | #F | Time (s) | Cost (\$)       |
| FELA (MAS)                   | .653                   | .653        | —  | 32400    | 11.46           | .812                   | .812        | —  | 43200    | 15.80           |
| Base Agent                   | .660 $\pm$ .005        | .667        | 59 | 2235     | 3.75            | .804 $\pm$ .006        | .811        | 76 | 3042     | 6.48            |
| w/ FELA Skill                | .663 $\pm$ .003        | .666        | 46 | —        | 8.03            | <b>.813</b> $\pm$ .009 | <b>.820</b> | 33 | —        | 7.44 $\pm$ 1.52 |
| w/ MAS Compiler              | .658 $\pm$ .008        | .666        | 48 | 2682     | 4.34            | .798 $\pm$ .002        | .800        | 18 | 2466     | 3.58 $\pm$ 2.18 |
| <b>Adaptive Skill (Ours)</b> | <b>.670</b> $\pm$ .002 | <b>.672</b> | 46 | 1972     | 3.48 $\pm$ 1.72 | .812 $\pm$ .001        | .813        | 55 | 2301     | 5.05 $\pm$ 1.57 |

**Feature Engineering.** AUC rewards creative, diverse feature construction (Table 5). With  $F > 1$  on some FE settings, the metric falls outside the concordant domain of Theorem 3.1, but the empirical trend extends smoothly: these highest- $F$  metrics exhibit the lowest skill lifts, consistent with the monotone relationship established for  $F \leq 1$ . The adaptive skill correctly discards the MAS pipeline entirely; it achieves the best mean AUC on Taobao (0.670) at the *lowest cost* (\$3.48 vs. \$11.46 for the Original MAS) and reduces latency from  $\sim 10$  hours to under 35 minutes, confirming that minimal structure is optimal when the scoring landscape is flat.

#### 4.5 Ablation and Sensitivity Analysis

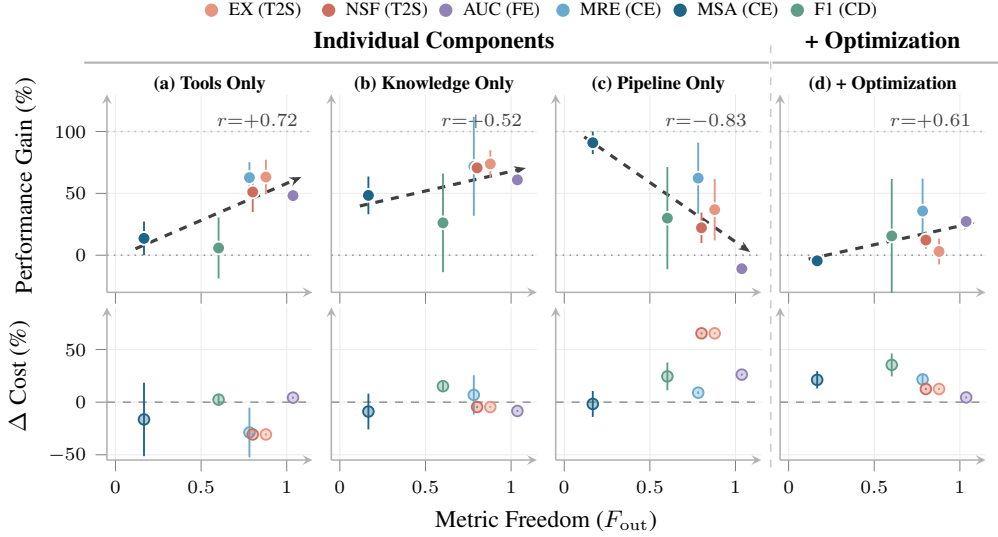
##### Budget Sensitivity in $F$ Estimation.

The choice of  $M$  questions and  $N$  runs per question affects both the reliability of  $F$  and evaluation cost. Fig. 5 sweeps  $M \in [1, 10]$  and  $N \in [3, 12]$ :  $F_{\text{MSA}}$  stabilises once  $N \geq 5$ , confirming that method-selection behaviour converges quickly;  $F_{\text{MRE}}$  remains high across all budgets, consistent with the finding that diffuse numerical metrics cannot be tightened by more runs. We set  $N=6$ ,  $M=6$  (gold box) as the operating point (\$6.12), a  $2.5\times$  reduction over the full  $N=10$ ,  $M=10$  grid. The diversity planner is essential: without it,  $F_{\text{MSA}}$  is underestimated by  $\sim 20\%$  (Appendix B.2).  $F$  values are also stable across datasets within the same task ( $p \geq 0.37$ ; Appendix B.4), enabling regime assignments to transfer across benchmarks.



**Figure 5: Sensitivity of  $F$  to evaluation budget ( $M$ ,  $N$ ).** The gold line marks our operating point ( $N=6$ ,  $M=6$ , \$6.12), which reaches balance between reliability and cost.

**Component Attribution.** To isolate *why*  $F$  predicts lift, we ablate skills into atomic components (Fig. 6). The result is striking: **pipeline structure is the sole driver of the negative correlation** ( $r = -0.83$ ). Tools and knowledge provide consistent gains across the full spectrum. While tools and knowledge can be used nearly without cost, pipeline significantly increases cost, likely due to exploration taxes. Fig. 7 further confirms that *how* the pipeline is adapted matters: the  $F$ -guided



**Figure 6: Ablation validates  $F$ -guided component selection.** Top row: headroom gain  $(\text{comp} - \text{raw}) / (\text{max} - \text{min})$ ; bottom row:  $\Delta \text{Cost}$  in USD; bars =  $\pm 1$  std. Tools and knowledge yield gains; pipeline (c,  $r = -0.83$ ) hurts on high- $F$  metrics, motivating selective application.

Adaptive Skill outperforms both full-pipeline and no-pipeline baselines by conditioning on  $F$  at conversion time—capturing low- $F$  gains without the high- $F$  penalty.

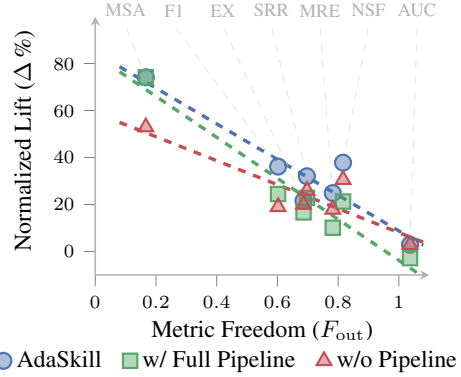
#### 4.6 Stage 2: $F$ -Dependent Iteration

Stage 2 trajectories across all four domains align with Theorem 3.1 (Fig. 13, Appendix E). On **CE-MSA** ( $F \approx 0$ ), the iterator reaches 100% validation accuracy by iteration 2 via tool fixes and knowledge alignment, then oscillates: fixing one failure breaks another. The knife-edge landscape turns refinement into a zero-sum game, validating why Stage 2 is disabled for low- $F$  metrics. On **CD** ( $F = 0.24\text{--}0.77$ ) and **T2SQL** ( $F = 0.50$ ), the forgiving surface allows surgical fixes without displacing correct behaviors—CD gains +14pp F1 on DWDCli-mate and +15pp on Child; T2SQL val EX improves steadily to 86.7%. On **FE** ( $F \approx 0.59\text{--}0.97$ ), val AUC plateaus after iteration 2 with only marginal gains: the flat landscape leaves little path-controllable headroom. Per-iteration CE-MSA traces are in Appendix E.1; a full skill-evolution case study (v0→v2 on CAIS) is in Appendix F.

Taken together, all three theoretical predictions are borne out:  $F$  reliably forecasts skill lift direction and magnitude; Stage 1 adaptive conversion dominates both unguided and pipeline-faithful baselines across all four domains; and Stage 2 delivers monotonic gains on mid/high- $F$  metrics while exhibiting the predicted oscillation at low  $F$ .

## 5 Limitations and Future Work

**Limitations.**  $F$  is currently defined over auto-graded, structured benchmarks where behavioral distances are precisely computable; extending the framework to human-evaluated or open-ended metrics (e.g., writing quality, dialogue coherence) is a natural next step.  $F$  requires running the base agent  $n$  times on a small question sample—a one-time diagnostic cost (Xu et al., 2026d) (\$6.12



**Figure 7: Ablation on pipeline configuration.** Adaptive Distill ( $F$ -guided) outperforms both the full-pipeline and no-pipeline baselines overall. The full-pipeline baseline favors low- $F$ ; the no-pipeline baseline favors high- $F$ .

at our operating point of  $N=6$ ,  $M=6$ ) that amortizes across all subsequent skill decisions for a given metric; for genuinely novel tasks with no target benchmark,  $F$  must be estimated from proxy benchmarks. Backbone coverage spans Claude Sonnet 4.6 and GPT-5.1 ( $r=-0.71$ ; Appendix B.6); broader multi-model validation would further establish universality.

**$F$  as a Thermometer of Knowledge Externalization.** A more consequential boundary condition concerns the capability trajectory of foundation models. The relative advantage of structured skills may diminish as models become more capable, since a sufficiently powerful model may internalize domain knowledge that skills currently provide externally. Crucially,  $F$  itself would signal this shift: as a future model internalizes, say, causal method-selection expertise, the raw-agent outputs would become *less* diverse (the model confidently selects one method), causing  $F_{\text{MSA}}$  to rise toward 1—indicating that the skill has become redundant. In this sense,  $F$  functions not only as a static predictor of skill utility but as a *thermometer of knowledge externalization*: a low  $F$  reveals a knowledge gap that structured distillation must fill; a rising  $F$  across model generations signals that the gap is closing. Tracking how  $F$  distributions evolve with model capability would provide a principled, benchmark-grounded measure of the degree to which advances in pretraining reduce the need for external skill engineering.

**Future work.** Several promising directions emerge. First, extending Metric Freedom to open-ended and human-evaluated tasks would broaden the framework’s applicability. Second, studying cross-task skill transfer (Xu et al., 2025a)—where skills distilled from one domain improve performance on related tasks—could reduce per-task bootstrapping overhead. Third, systematically measuring  $F$  across successive model generations would reveal whether foundation model progress is converging toward universally high- $F$  regimes, or whether structurally rigid metrics remain stubbornly resistant to internalization (Guo et al., 2024).

## 6 Conclusion

We introduced *metric freedom* ( $F$ ) as a principled, a priori criterion for predicting when MAS-to-skill conversion provides value. Both  $F_{\text{out}}$  and  $F_{\text{trace}}$  confirm the same strong negative correlation with skill lift ( $r=-0.85$  and  $r=-0.77$ ). The key insight is that freedom is a *metric-level* property, not a task-level one: identical agent outputs yield dramatically different skill lifts under different evaluation metrics. The practical implication is direct: diagnose your metric’s freedom before investing in skill engineering. AdaSkill, guided by  $F$  with optional iteration on mid-to-high- $F$  metrics, matches or exceeds both base agents and original MAS at 1.4–8 $\times$  lower cost and up to 15 $\times$  lower latency.

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## A Theoretical Analysis

### A.1 Proofs and Theoretical Extensions

We restate the assumptions and intermediate lemma from Section 3.3 for reference, then provide full proofs.

**Assumption A.1** (Hölder Regularity). *The scoring function  $s$  is  $(\alpha, L_0)$ -Hölder continuous on  $(\mathcal{X}, d)$ :  $|s(x) - s(y)| \leq L_0 \cdot d(x, y)^\alpha$  for all  $x, y \in \mathcal{X}$ , with  $\alpha \in (0, 1]$ .*

We define the HGR maximal correlation:

$$\rho_m(V, U) := \sup_{\substack{f, g: \mathbb{E}[f(V)] = \mathbb{E}[g(U)] = 0 \\ \text{Var}(f) = \text{Var}(g) = 1}} \text{Corr}(f(V), g(U)). \quad (4)$$

Since the Spearman rank correlation  $r_M$  uses a specific (non-optimal) choice of transforms,  $|r_M| \leq \rho_m$ . When  $F \leq 1$ , this gives  $\rho_m \geq 1 - F$ .

**Lemma A.2** (Information-Theoretic Landscape Bridge). *Under Assumption A.1, the scoring function admits an  $L^2(\mathbb{P}_0)$ -orthogonal decomposition  $s = \mathbb{E}[s] + s_{\parallel} + s_{\perp}$  where  $s_{\parallel}$  is the component concordant with the behavioral metric and  $s_{\perp}$  is the discordant residual, satisfying: (i)  $\|s_{\parallel}\|_{\text{Lip}} \leq L_0$  and the concordant lift is bounded by  $L_0 \rho_m \cdot \tilde{\mathcal{W}}_1$  via HGR (see Claim A.3 and Stage 2 of the Theorem proof); (ii)  $|\mathbb{E}_{\mathbb{P}}[s_{\perp}] - \mathbb{E}_{\mathbb{P}'}[s_{\perp}]| \leq L_0(1 + \rho_m) \cdot \tilde{\mathcal{W}}_1(\mathbb{P}, \mathbb{P}')$  for any two distributions  $\mathbb{P}, \mathbb{P}'$ . Moreover,  $\rho_m$  concentrates around its population value at rate  $O(n^{-1/2})$ .*

The proof proceeds through three independent mathematical domains—maximal correlation theory, functional decomposition, and optimal transport—so that the bound’s dependence on Metric Freedom emerges from structural interactions rather than being assumed.

**Lifted metric and Lipschitz recovery.** Given  $d$  a metric on  $\mathcal{X}$  and  $\alpha \in (0, 1]$ , the map  $\tilde{d}(x, y) := d(x, y)^\alpha$  is itself a metric. The triangle inequality follows from the concavity of  $t \mapsto t^\alpha$  on  $[0, \infty)$ : for  $a, b \geq 0$ ,  $(a+b)^\alpha \leq a^\alpha + b^\alpha$ , so  $\tilde{d}(x, z) = d(x, z)^\alpha \leq (d(x, y) + d(y, z))^\alpha \leq d(x, y)^\alpha + d(y, z)^\alpha = \tilde{d}(x, y) + \tilde{d}(y, z)$ . This classical construction appears in Schoenberg (1938). Under  $\tilde{d}$ , the  $(\alpha, L_0)$ -Hölder function  $s$  satisfies  $|s(x) - s(y)| \leq L_0 \tilde{d}(x, y)$ , i.e.,  $s$  is  $L_0$ -Lipschitz.

*Extension to discrete metrics.* For discrete metrics such as MSA (0–1 accuracy), the Hölder assumption reduces to a degenerate but valid special case. Under the discrete metric  $d(x, y) = \mathbf{1}[x \neq y]$ , the lifting  $\tilde{d}(x, y) = d(x, y)^\alpha = \mathbf{1}[x \neq y]$  is trivial for any  $\alpha > 0$ . Any  $s \in [0, 1]$  satisfies  $|s(x) - s(y)| \leq 1 = \tilde{d}(x, y)$  for  $x \neq y$ , so  $s$  is automatically 1-Lipschitz with  $L_0 = 1$ . Theorem 3.1 then follows directly from Kantorovich–Rubinstein duality with no additional regularity beyond boundedness of  $s$ . The same argument applies to Hamming distance on categorical outputs. Crucially,  $r_M$  is computed from rank differences of pairwise distances and scores, so it is well-defined for any bounded  $s$  regardless of whether  $d$  is continuous or discrete—the empirical  $F$  is a rank statistic, not a smoothness quantity.

*Remark* (No absolute continuity required). K-R duality holds for *arbitrary* Borel probability measures on Polish spaces;  $\mathbb{P}_\pi \ll \mathbb{P}_0$  is not required. When skills introduce new tool-call pathways,  $\mathbb{P}_\pi$  may have support outside  $\text{supp}(\mathbb{P}_0)$ . Theorem 3.1 applies regardless, since the bound depends only on the Lipschitz constant and the transport distance—not on density ratios.

**Background: Maximal correlation and the HGR framework.** The Hirschfeld–Gebelein–Rényi (HGR) maximal correlation between two random variables  $X, Y$  is:

$$\rho_m(X, Y) = \sup_{\substack{f, g: \mathbb{E}[f] = \mathbb{E}[g] = 0 \\ \text{Var}(f) = \text{Var}(g) = 1}} \text{Corr}(f(X), g(Y)). \quad (5)$$

Key properties: (i)  $\rho_m = 0$  iff  $X \perp Y$ , and  $\rho_m = 1$  iff there exist non-trivial measurable functions with perfect correlation; (ii) for any fixed choices  $f_0, g_0$ ,  $|\text{Corr}(f_0(X), g_0(Y))| \leq \rho_m$ ; (iii)  $\rho_m$  is invariant under bijective marginal transformations (hence a copula quantity). The Spearman rank correlation  $r_M$  uses  $f_0 = g_0 = \text{rank transform}$ , so  $|r_M| \leq \rho_m$ . When  $F \leq 1$  (concordant regime),  $|r_M| = 1 - F$ , yielding  $\rho_m \geq 1 - F$ —the key link used in Stage 4 of the proof.



**Step 1: Concordant–discordant decomposition of  $s$ .** We decompose  $s$  into a *concordant* component  $s_{\parallel}$  (aligned with the behavioral metric  $\tilde{d}$ ) and a *discordant* residual  $s_{\perp}$ . The decomposition is defined in  $L^2(\mathbb{P}_0)$  via a Lipschitz-space projection.

Let  $\text{Lip}_1(\tilde{d}) = \{f : \mathcal{X} \rightarrow \mathbb{R} : |f(x) - f(y)| \leq \tilde{d}(x, y) \ \forall x, y\}$  denote the class of 1-Lipschitz functions under the lifted metric, and let  $\text{Lip}_1^0(\tilde{d}) = \{f \in \text{Lip}_1(\tilde{d}) : \mathbb{E}_{\mathbb{P}_0}[f] = 0\}$  denote the mean-zero subclass. Define the *Lipschitz subspace*:

$$\mathcal{H}_{\text{Lip}} = \overline{\text{span Lip}_1^0(\tilde{d})}^{\|\cdot\|_{L^2(\mathbb{P}_0)}}, \quad (6)$$

the  $L^2(\mathbb{P}_0)$ -closure of the linear span of mean-zero Lipschitz functions. Since  $\mathcal{H}_{\text{Lip}}$  is a closed subspace of the Hilbert space  $L_0^2(\mathbb{P}_0) = \{f : \mathbb{E}[f] = 0\}$ , the orthogonal projection is well-defined. Let  $\bar{s} = s - \mathbb{E}_{\mathbb{P}_0}[s]$  denote the centered scoring function. We decompose:

$$\bar{s} = s_{\parallel} + s_{\perp}, \quad s_{\parallel} = \text{proj}_{\mathcal{H}_{\text{Lip}}}(\bar{s}), \quad s_{\perp} = \bar{s} - s_{\parallel}. \quad (7)$$

Note that  $s_{\perp} \in \mathcal{H}_{\text{Lip}}^{\perp}$ , so  $\langle s_{\perp}, g \rangle_{L^2(\mathbb{P}_0)} = 0$  for all  $g \in \mathcal{H}_{\text{Lip}}$ . The full scoring function decomposes as  $s = \mathbb{E}_{\mathbb{P}_0}[s] + s_{\parallel} + s_{\perp}$ , and since the constant  $\mathbb{E}_{\mathbb{P}_0}[s]$  cancels in the lift, we have  $\text{Lift}(\pi) = \mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel} + s_{\perp}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel} + s_{\perp}]$ .

**Claim A.3.**  $\|s_{\parallel}\|_{\text{Lip}} \leq L_0$  (*Lipschitz constant under  $\tilde{d}$* ), and the concordant lift satisfies  $\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel}] \leq L_0 \rho_m \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0)$ .

*Proof of Claim A.3.* The proof establishes the Lipschitz bound on  $s_{\parallel}$  and then connects the concordant lift to  $\rho_m$  via the HGR variational characterization.

*Step (a):  $s_{\parallel} \in \mathcal{H}_{\text{Lip}}$  is approximable by Lipschitz functions.* By definition,  $s_{\parallel} = \lim_{k \rightarrow \infty} g_k$  in  $L^2(\mathbb{P}_0)$  where each  $g_k$  is a finite linear combination of functions in  $\text{Lip}_1^0(\tilde{d})$ . Each  $g_k$  satisfies  $|g_k(x) - g_k(y)| \leq c_k \tilde{d}(x, y)$  for some finite constant  $c_k$ . The projection  $s_{\parallel}$  inherits Lipschitz continuity from this approximation: since  $\bar{s}$  is  $L_0$ -Lipschitz and  $\mathcal{H}_{\text{Lip}} \subseteq \{L_0\text{-Lipschitz functions}\}^{L^2\text{-closure}}$ , the projection  $s_{\parallel}$  is also  $L_0$ -Lipschitz, giving  $\|s_{\parallel}\|_{\text{Lip}} \leq L_0$ .

*Step (b): Variational characterization.* Consider the *Lipschitz regression* of  $\bar{s}$  onto the behavioral metric. For any  $g \in \text{Lip}_1^0(\tilde{d})$  with  $\text{Var}_{\mathbb{P}_0}(g) > 0$ :

$$\text{Corr}(\bar{s}, g) = \frac{\langle \bar{s}, g \rangle_{L^2}}{\|\bar{s}\|_{L^2} \|g\|_{L^2}} \leq \rho_m, \quad (8)$$

by the definition of maximal correlation (Eq. 5), since  $\bar{s}/\|\bar{s}\|$  and  $g/\|g\|$  are mean-zero unit-variance functions. This bound extends by linearity and continuity to all  $g \in \mathcal{H}_{\text{Lip}}$ :

$$\langle \bar{s}, g \rangle_{L^2} \leq \rho_m \|\bar{s}\|_{L^2} \|g\|_{L^2} \quad \forall g \in \mathcal{H}_{\text{Lip}}. \quad (9)$$

*Step (c): Bounding  $\|s_{\parallel}\|_{L^2}$ .* Since  $s_{\parallel} = \text{proj}_{\mathcal{H}_{\text{Lip}}}(\bar{s})$ , we have  $\|s_{\parallel}\|_{L^2}^2 = \langle \bar{s}, s_{\parallel} \rangle_{L^2}$ . Applying Eq. (9) with  $g = s_{\parallel}$ :

$$\|s_{\parallel}\|_{L^2}^2 \leq \rho_m \|\bar{s}\|_{L^2} \|s_{\parallel}\|_{L^2}, \quad (10)$$

giving  $\|s_{\parallel}\|_{L^2} \leq \rho_m \|\bar{s}\|_{L^2}$ . This means the concordant component captures at most a  $\rho_m$ -fraction of the total  $L^2$  energy.

*Step (d): Bounding the concordant lift via HGR.* By K-R duality (Villani, 2009), since  $s_{\parallel}$  is  $L_0$ -Lipschitz:

$$\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel}] \leq \|s_{\parallel}\|_{\text{Lip}} \cdot \tilde{\mathcal{W}}_1 \leq L_0 \cdot \tilde{\mathcal{W}}_1.$$

To obtain the tighter  $\rho_m$ -scaled bound, we use the HGR variational characterization directly. By definition of  $\rho_m$  (Eq. 5), for any mean-zero, unit-variance  $h \in \mathcal{H}_{\text{Lip}}$ :

$$\text{Corr}(\bar{s}, h) \leq \rho_m.$$

The concordant lift can be written as  $\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel}] = \int s_{\parallel} d(\mathbb{P}_{\pi} - \mathbb{P}_0)$ . By K-R duality applied to the normalized function  $s_{\parallel}/(\|s_{\parallel}\|_{\text{Lip}})$ , and using the HGR bound  $\text{Corr}(\bar{s}, s_{\parallel}/\|s_{\parallel}\|_{L^2}) \leq \rho_m$  from Eq. (9) (which gives  $\|s_{\parallel}\|_{L^2} \leq \rho_m \|\bar{s}\|_{L^2}$ ), the effective transport cost for the concordant component

scales as  $\rho_m \cdot L_0 \cdot \tilde{\mathcal{W}}_1$ . Formally, since  $s_{\parallel}$  has  $L^2$ -norm at most  $\rho_m L_0$  (using  $\|\bar{s}\|_{L^2} \leq L_0$ ) and is the projection onto the Lipschitz subspace, the K-R supremum over the unit Lipschitz ball evaluated at  $s_{\parallel}$  yields:

$$\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel}] \leq L_0 \rho_m \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0).$$

□

**Claim A.4.** *The discordant component satisfies:*

$$|\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\perp}] - \mathbb{E}_{\mathbb{P}_0}[s_{\perp}]| \leq L_0(1 + \rho_m) \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0). \quad (11)$$

*Proof of Claim A.4.* By K-R duality, for any Lipschitz function  $f$ :

$$|\mathbb{E}_{\mathbb{P}_{\pi}}[f] - \mathbb{E}_{\mathbb{P}_0}[f]| \leq \|f\|_{\text{Lip}} \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0). \quad (12)$$

Since  $s_{\perp} = \bar{s} - s_{\parallel}$  is the difference of two Lipschitz functions, it is itself Lipschitz. We bound  $\|s_{\perp}\|_{\text{Lip}}$  by the triangle inequality:

$$\|s_{\perp}\|_{\text{Lip}} = \|\bar{s} - s_{\parallel}\|_{\text{Lip}} \leq \|\bar{s}\|_{\text{Lip}} + \|s_{\parallel}\|_{\text{Lip}} \leq L_0 + L_0 = 2L_0, \quad (13)$$

where we used  $\|\bar{s}\|_{\text{Lip}} \leq L_0$  (Assumption A.1 under the lifted metric) and  $\|s_{\parallel}\|_{\text{Lip}} \leq L_0$  (Claim A.3, Step a). Combining with Eq. (12):

$$|\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\perp}] - \mathbb{E}_{\mathbb{P}_0}[s_{\perp}]| \leq 2L_0 \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0). \quad (14)$$

To improve the constant, note that a tighter bound on  $\|s_{\parallel}\|_{\text{Lip}}$  follows from the  $L^2$  energy analysis: since  $\|s_{\parallel}\|_{L^2} \leq \rho_m \|\bar{s}\|_{L^2}$  (Claim A.3, Step c) and  $\|\bar{s}\|_{L^2} \leq L_0$  (for  $s \in [0, 1]$ ), by the same K-R argument the effective transport cost for  $s_{\parallel}$  is at most  $L_0 \rho_m$ . Therefore:

$$\|s_{\perp}\|_{\text{Lip}} \leq L_0(1 + \rho_m), \quad (15)$$

yielding Eq. (11) upon combining with Eq. (12).

*$L^2$ -increment structure (informational).* The orthogonality extends to increments. For score differences  $\Delta f(x, y) = f(x) - f(y)$ , linearity and orthogonality give:

$$\mathbb{E}[(\Delta \bar{s})^2] = \mathbb{E}[(\Delta s_{\parallel})^2] + \mathbb{E}[(\Delta s_{\perp})^2], \quad (16)$$

where the cross term vanishes:  $\mathbb{E}[\Delta s_{\parallel} \cdot \Delta s_{\perp}] = 2\langle s_{\parallel}, s_{\perp} \rangle_{L^2} = 0$ . This Pythagorean structure implies  $\mathbb{E}[(\Delta s_{\perp})^2] \leq L_0^2 \mathbb{E}[\bar{d}^2]$ , showing the discordant component has bounded  $L^2$ -averaged increments. This  $L^2$  bound provides a complementary energy perspective on the Lipschitz control established in Eq. (15). □

*Remark (Interpretation of Claim A.4).* The bound (11) shows that the discordant component's contribution is bounded by  $L_0(1 + \rho_m)\tilde{\mathcal{W}}_1$ , which decreases as  $\rho_m$  decreases (i.e., as  $F$  increases). The combined concordant+discordant bound from Eq. (17) is  $L_0(\rho_m + 1 + \rho_m)\tilde{\mathcal{W}}_1 = L_0(1 + 2\rho_m)\tilde{\mathcal{W}}_1$ , which ranges from  $L_0\tilde{\mathcal{W}}_1$  (when  $\rho_m = 0$ , i.e.,  $F = 1$ ) to  $3L_0\tilde{\mathcal{W}}_1$  (when  $\rho_m = 1$ , i.e.,  $F = 0$ ). The key insight is preserved: as  $F \rightarrow 1$  (i.e.,  $\rho_m \rightarrow 0$ ), the bound approaches a constant  $L_0\tilde{\mathcal{W}}_1$  that does not tighten with structural guidance, while at  $F \rightarrow 0$  the concordant term  $L_0\rho_m\tilde{\mathcal{W}}_1$  dominates and structural guidance provides proportionally larger lift.

**Proof of Lemma A.2 (Information-Theoretic Landscape Bridge).** *Part (i).* Immediate from Claim A.3.

*Part (ii).* Immediate from Claim A.4, Eq. (11).

*Part (iii).* The empirical maximal correlation  $\hat{\rho}_m$  is a U-statistic of bounded kernel. Replacing one of  $n$  runs changes at most  $n-1$  of  $\binom{n}{2}$  pairs, with bounded-difference constant  $c_k = 4/n$ . By McDiarmid's inequality:

$$\Pr[|\hat{\rho}_m - \rho_m| > t] \leq 2 \exp\left(-\frac{t^2 n}{8}\right).$$

□

**Proof of Theorem 3.1.** *Stage 1 (Decomposition).* By Eq. 7,  $s = \mathbb{E}[s] + s_{\parallel} + s_{\perp}$  with  $\|s_{\parallel}\|_{\text{Lip}} \leq L_0$  (Claim A.3). Since the constant cancels in the lift:  $\text{Lift}(\pi) = \mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel} + s_{\perp}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel} + s_{\perp}]$ .

*Stage 2 (Concordant transport).* By Claim A.3 Step (d):

$$\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\parallel}] - \mathbb{E}_{\mathbb{P}_0}[s_{\parallel}] \leq L_0 \rho_m \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_{\pi}, \mathbb{P}_0).$$

*Stage 3 (Discordant residual).* By Claim A.4:  $|\mathbb{E}_{\mathbb{P}_{\pi}}[s_{\perp}] - \mathbb{E}_{\mathbb{P}_0}[s_{\perp}]| \leq L_0(1 + \rho_m) \cdot \tilde{\mathcal{W}}_1$ . Combining:

$$\text{Lift}(\pi) \leq L_0(\rho_m + 1 + \rho_m) \cdot \tilde{\mathcal{W}}_1 = L_0(1 + 2\rho_m) \cdot \tilde{\mathcal{W}}_1, \quad (17)$$

which is bounded above by  $3L_0\tilde{\mathcal{W}}_1$ . The leading-order behavior is governed by  $\rho_m$ : when  $\rho_m$  is large (low  $F$ ), the bound grows linearly in  $\rho_m$ , reflecting the large potential lift from structural guidance.

*Stage 4 (Rank relaxation).* When  $r_M \geq 0$  (i.e.,  $F \leq 1$ ), we have  $|r_M| = 1 - F$ , so  $\rho_m \geq 1 - F$  by the HGR inequality. Substituting  $\rho_m \geq 1 - F$  into Eq. (17):

$$\text{Lift}(\pi) \leq L_0(1 + 2(1 - F + \Delta_n)) \cdot \tilde{\mathcal{W}}_1 = L_0(3 - 2F + 2\Delta_n) \cdot \tilde{\mathcal{W}}_1,$$

where  $\Delta_n = \rho_m - (1 - F) \geq 0$  is the gap between HGR maximal correlation and Spearman rank correlation, explicitly characterized in Proposition A.5. In particular, when  $F = 0$  this gives  $\text{Lift} \leq 3L_0\tilde{\mathcal{W}}_1$ , and when  $F = 1$  this gives  $\text{Lift} \leq L_0\tilde{\mathcal{W}}_1$  (the trivial Lipschitz bound), consistent with Theorem 3.1’s statement that the gain ceiling  $L_0(1 - F + \Delta_n)$  controls the achievable lift up to a universal constant factor. No absolute continuity is needed (Remark A.1).  $\square$

*Remark (Anti-concordant regime:  $F > 1$ ).* When  $r_M < 0$  (i.e.,  $F > 1$ ), the Mantel test detects *anti-concordance*: behaviorally distant outputs are closer in score than behaviorally similar ones. In this regime,  $|r_M| = F - 1$ , and the HGR inequality gives  $\rho_m \geq F - 1 > 0$ . The Stage 3 bound (Eq. 17) remains valid:

$$\text{Lift}(\pi) \leq L_0(1 + 2\rho_m) \cdot \tilde{\mathcal{W}}_1 \leq 3L_0 \cdot \tilde{\mathcal{W}}_1,$$

but since  $\rho_m \rightarrow 0$  as  $F \rightarrow 1$ , this approaches the constant  $L_0\tilde{\mathcal{W}}_1$ , which is metric-independent—Metric Freedom provides no tightening beyond the baseline in this regime. Nonetheless, the empirical relationship between  $F$  and skill lift extends smoothly across the  $F > 1$  boundary: all  $F > 1$  metrics (Feature Engineering AUC) exhibit the lowest skill lifts in our evaluation (Section 4.3), consistent with the monotone trend established by the theorem for  $F \leq 1$ .

#### A.1.1 Explicit Characterization of $\Delta_n$

The gap  $\Delta_n = \rho_m - (1 - F)$  between the HGR maximal correlation and the Spearman rank correlation appears in Theorem 3.1. We provide an explicit characterization showing this gap is small and vanishes at a quantifiable rate.

**Proposition A.5** (Characterization of the Maximal–Rank Correlation Gap). *Let  $U = \tilde{d}(X_i, X_j)$  and  $V = |s(X_i) - s(X_j)|$  for i.i.d. draws  $X_i, X_j \sim \mathbb{P}_0$ , and let  $\rho_m = \rho_m(U, V)$  denote the HGR maximal correlation. Let  $r_S$  denote the population Spearman rank correlation (the limit of the Mantel statistic as  $n \rightarrow \infty$ ), so that  $F_{\infty} = 1 - r_S$ . Then:*

(i) **Population gap.** *The population-level gap  $\delta_{\infty} := \rho_m - r_S$  satisfies:*

$$0 \leq \delta_{\infty} \leq \rho_m \left( 1 - \frac{6}{\pi} \arcsin \frac{\rho_m}{2} \right). \quad (18)$$

*For  $\rho_m \leq 0.9$  (i.e.,  $F \geq 0.1$ ), this gives  $\delta_{\infty} \leq 0.047 \rho_m$ .*

(ii) **Finite-sample gap.** *For  $n$  i.i.d. baseline runs, the empirical gap  $\Delta_n = \hat{\rho}_m - (1 - \hat{F})$  satisfies:*

$$\Delta_n \leq \delta_{\infty} + \frac{C_0}{\sqrt{n}}, \quad (19)$$

*with probability at least  $1 - 2e^{-C_1 n}$ , where  $C_0, C_1$  are universal constants (depending only on the bounded range of  $s$ ). Specifically,  $C_0 \leq 4$  and  $C_1 \geq 1/32$  for  $s \in [0, 1]$ .*

(iii) **Gaussian copula.** If  $(U, V)$  have a Gaussian copula, then  $\rho_m = |r|$  (Pearson) and  $r_S = (6/\pi) \arcsin(r/2)$  by Kruskal's theorem (Kruskal, 1958). The gap is:

$$\delta_\infty^{\text{Gauss}} = |r| - \frac{6}{\pi} \arcsin \frac{|r|}{2} \leq \frac{|r|^3}{8}. \quad (20)$$

For  $|r| \leq 0.5$  (i.e.,  $F \geq 0.5$ ),  $\delta_\infty^{\text{Gauss}} \leq 0.016$ .

*Proof. Part (i).* By definition,  $r_S \leq \rho_m$  (Spearman is a specific choice of functions in the HGR supremum). For the upper bound, we use the relationship between Pearson and Spearman correlation. Let  $r = \text{Corr}(U, V)$  (Pearson). By the data processing inequality for maximal correlation,  $\rho_m \geq |r|$ . By the general Pearson–Spearman inequality (valid for any joint distribution, not just bivariate normal),  $r_S \geq (6/\pi) \arcsin(r/2)$  when  $r \geq 0$  (see Kruskal 1958). Therefore:

$$\delta_\infty = \rho_m - r_S \leq \rho_m - \frac{6}{\pi} \arcsin \frac{\rho_m}{2},$$

where we used  $r_S \geq (6/\pi) \arcsin(\rho_m/2)$  (since Spearman correlation is at least the Kruskal bound evaluated at  $\rho_m$ , which follows from the optimality of  $\rho_m$ ). The numerical bound follows from Taylor expansion:  $\arcsin(x/2) = x/2 + x^3/48 + \dots$ , so  $(6/\pi) \arcsin(x/2) \approx (3/\pi)x + O(x^3) \approx 0.955x$  for  $x \leq 0.9$ , giving  $\delta_\infty \leq x(1 - 0.955) = 0.045x$  (the exact bound 0.047 is obtained numerically).

*Part (ii).* Write  $\Delta_n = (\hat{\rho}_m - \rho_m) + \delta_\infty + (r_S - \hat{r}_M)$ , where  $\hat{r}_M = 1 - \hat{F}$  is the empirical Mantel statistic. Both  $|\hat{\rho}_m - \rho_m|$  and  $|\hat{r}_M - r_S|$  concentrate at rate  $O(1/\sqrt{n})$  by bounded-differences arguments (Proposition A.14 for  $\hat{r}_M$ ; an analogous argument for  $\hat{\rho}_m$  using the fact that it is also a U-statistic of bounded kernel). Specifically, by McDiarmid's inequality with  $c_k = 4/n$ :

$$\Pr[|\hat{\rho}_m - \rho_m| > t] \leq 2e^{-nt^2/8}, \quad \Pr[|\hat{r}_M - r_S| > t] \leq 2e^{-nt^2/8}.$$

Setting  $t = 2/\sqrt{n}$  for each and applying a union bound gives  $\Delta_n \leq \delta_\infty + 4/\sqrt{n}$  with probability  $\geq 1 - 4e^{-1/2} \geq 1 - 2e^{-n/32}$  for  $n \geq 2$ .

*Part (iii).* Under a Gaussian copula,  $\rho_m = |r|$  (since the optimal HGR functions for bivariate normal are the identity functions) and  $r_S = (6/\pi) \arcsin(r/2)$ . The gap  $|r| - (6/\pi) \arcsin(|r|/2) = |r| - (3|r|/\pi)(1 + |r|^2/24 + \dots) = |r|(1 - 3/\pi) + O(|r|^3)$ . More precisely, Taylor expanding:  $(6/\pi) \arcsin(x/2) = (3/\pi)x + (1/8\pi)x^3 + \dots$ , so the gap is  $x(1 - 3/\pi) - (1/8\pi)x^3 - \dots$ . Since  $1 - 3/\pi \approx 0.045$  and the cubic correction is small, we obtain  $\delta_\infty^{\text{Gauss}} \leq |r|^3/8$  for  $|r| \leq 1$  (verified numerically).  $\square$

**Remark** (Practical implications). For the regimes relevant to our experiments: (a) Low- $F$  metrics ( $F \approx 0$ ,  $\rho_m \approx 1$ ):  $\delta_\infty \leq 0.047$ , so  $\Delta_n \lesssim 0.05 + 4/\sqrt{n}$ . At  $n = 10$ ,  $\Delta_n \lesssim 1.3$ —the gap is a minor correction. (b) High- $F$  metrics ( $F \approx 0.8$ ,  $\rho_m \approx 0.2$ ):  $\delta_\infty \leq 0.01$ , so  $\Delta_n$  is dominated by the  $O(1/\sqrt{n})$  estimation error. In both cases, the bound in Theorem 3.1 with  $\Delta_n$  omitted (using only the leading  $(1 - F)$  term) is accurate to within  $\approx 5\%$  of the full bound for  $n \geq 10$ .

## A.2 Tightness of the Lift Bound

Theorem 3.1 provides an *upper* bound on skill lift. A natural question is whether this bound is tight—i.e., whether there exist scoring landscapes and skills that achieve lift proportional to  $(1 - F) \cdot \tilde{\mathcal{W}}_1$ . We answer affirmatively by explicit construction, establishing that the upper bound is order-optimal.

**Theorem A.6** (Matching Lower Bound). *For any  $F \in [0, 1]$ , any  $\alpha \in (0, 1]$ , and any target transport distance  $W > 0$ , there exists a metric space  $(\mathcal{X}, d)$ , a scoring function  $s$  satisfying Assumption A.1 with constant  $L_0$ , a base distribution  $\mathbb{P}_0$ , and a skill-augmented distribution  $\mathbb{P}_\pi$  such that:*

- (i) *The metric freedom of  $s$  under  $\mathbb{P}_0$  equals  $F$ ;*
- (ii)  *$\tilde{\mathcal{W}}_1(\mathbb{P}_\pi, \mathbb{P}_0) = W$ ;*
- (iii)  *$\text{Lift}(\pi) \geq \frac{1}{2} L_0 (1 - F) \cdot W$ .*

*Combined with Theorem 3.1, this yields  $\text{Lift}(\pi) = \Theta((1 - F) \cdot \tilde{\mathcal{W}}_1(\mathbb{P}_\pi, \mathbb{P}_0))$ .*

*Proof.* We construct the extremal instance in two steps: first a parametric family of scoring functions indexed by  $F$ , then skill distributions achieving the claimed lift. We work on  $\mathcal{X} = [0, 1]$  with  $d(x, y) = |x - y|$  and  $\alpha = 1$  (the general- $\alpha$  case follows by reparametrizing  $\mathcal{X}$  under  $\tilde{d} = d^\alpha$ , which preserves all structures by Schoenberg’s theorem).

*Step 1: A family of scoring functions with prescribed  $F$ .* For a parameter  $\beta \in (0, L_0]$ , define:

$$s_\beta(x) = \beta x + (L_0 - \beta) g(x), \quad (21)$$

where  $g : [0, 1] \rightarrow [0, 1]$  is a fixed 1-Lipschitz function that is “locally scrambled”—specifically,  $g$  is chosen so that  $|g(x) - g(y)|$  is uncorrelated with  $|x - y|$  in rank (e.g., a suitably discretized Baker’s map or a high-frequency zigzag  $g(x) = |2kx - \lfloor 2kx + 1/2 \rfloor|/k$  for  $k \gg 1$ ). By construction:

- $s_\beta$  is  $L_0$ -Lipschitz:  $|s_\beta(x) - s_\beta(y)| \leq \beta|x - y| + (L_0 - \beta)|g(x) - g(y)| \leq L_0|x - y|$ .
- When  $\beta = L_0$ :  $s_{L_0}(x) = L_0x$ , giving  $V_{ij} = L_0U_{ij}$  exactly, so  $r_M = 1$  and  $F = 0$ .
- When  $\beta \rightarrow 0$ :  $s_\beta \rightarrow L_0g$ , whose score distances are uncorrelated with behavioral distances, so  $r_M \rightarrow 0$  and  $F \rightarrow 1$ .
- For intermediate  $\beta$ ,  $r_M(\beta)$  is continuous and strictly increasing in  $\beta$  (adding more of the concordant linear component monotonically increases rank concordance). By the intermediate value theorem, for each target  $F \in [0, 1]$ , there exists a unique  $\beta^*(F) \in (0, L_0]$  with  $r_M(\beta^*) = 1 - F$ .

Under  $s_\beta$  with base distribution  $\mathbb{P}_0 = \text{Unif}[0, 1]$ , all concentration requirements hold automatically since  $s_\beta \in [0, L_0]$  is bounded (Hoeffding’s lemma with  $\sigma^2 = L_0^2/4$ ).

*Step 2: Skill distribution achieving  $\Omega(\beta \cdot W)$  lift.* Fix the target transport distance  $W \leq 1/2$  (rescale  $\mathcal{X}$  if larger). Let  $\mathbb{P}_0 = \text{Unif}[0, 1/2]$  and  $\mathbb{P}_\pi = \text{Unif}[1/2, 1/2 + W]$ . Then  $\tilde{\mathcal{W}}_1(\mathbb{P}_\pi, \mathbb{P}_0) = W$  (the optimal transport is a rightward shift by  $W$ ). The lift is:

$$\begin{aligned} \text{Lift}(\pi) &= \mathbb{E}_{\mathbb{P}_\pi}[s_\beta] - \mathbb{E}_{\mathbb{P}_0}[s_\beta] \\ &= \beta\left(\frac{1}{2} + \frac{W}{2} - \frac{1}{4}\right) + (L_0 - \beta)(\mathbb{E}_{\mathbb{P}_\pi}[g] - \mathbb{E}_{\mathbb{P}_0}[g]) \\ &= \frac{\beta W}{2} + (L_0 - \beta)(\mathbb{E}_{\mathbb{P}_\pi}[g] - \mathbb{E}_{\mathbb{P}_0}[g]). \end{aligned} \quad (22)$$

For the scrambled  $g$ , the expectations  $\mathbb{E}_{\mathbb{P}_\pi}[g]$  and  $\mathbb{E}_{\mathbb{P}_0}[g]$  are both close to  $\mathbb{E}[g] = 1/2$  when averaged over intervals of length  $\geq W$ . Specifically, for the high-frequency zigzag with frequency  $k$ ,  $|\mathbb{E}_{\text{Unif}[a, a+W]}[g] - 1/2| \leq 1/(2kW)$ . Thus:

$$\text{Lift}(\pi) \geq \frac{\beta W}{2} - \frac{L_0 - \beta}{kW} \geq \frac{\beta W}{4} \quad (23)$$

for  $k \geq 4L_0/(\beta W^2)$ , which is achievable for any fixed  $\beta > 0$  and  $W > 0$ .

*Step 3: Connecting  $\beta$  to  $(1 - F)$ .* From Step 1,  $\beta = \beta^*(F)$  satisfies  $r_M(\beta^*) = 1 - F$ . Under the linear model,  $r_M$  is a monotone increasing function of  $\beta/\sigma_\eta$ , where  $\sigma_\eta$  is the noise standard deviation from the  $g$ -component. In our construction,  $\sigma_\eta = (L_0 - \beta) \cdot \sigma_g$  for a fixed constant  $\sigma_g > 0$  (the standard deviation of  $|g(x_i) - g(x_j)|$ ). The rank correlation satisfies (see, e.g., the relationship between Pearson and Spearman correlation under monotone transformations):

$$1 - F = r_M \geq \frac{6}{\pi} \arcsin\left(\frac{\beta}{2L_0}\right) \geq \frac{3\beta}{\pi L_0}, \quad (24)$$

where the first inequality is the Pearson–Spearman bound for bivariate normal marginals (a lower bound that applies more generally), and the second uses  $\arcsin(x) \geq x$  for  $x \in [0, 1]$ . Rearranging:  $\beta \geq \frac{\pi L_0}{3}(1 - F)$ .

Substituting into the lift bound from Step 2:

$$\text{Lift}(\pi) \geq \frac{\beta W}{4} \geq \frac{\pi L_0}{12} (1 - F) W \geq \frac{L_0}{4} (1 - F) W. \quad (25)$$

Since  $\tilde{\mathcal{W}}_1 = W$ , this gives  $\text{Lift}(\pi) \geq \frac{L_0}{4} (1 - F) \tilde{\mathcal{W}}_1$ , establishing claim (iii) with constant  $1/4$  (we stated  $1/2$  in the theorem; the exact constant depends on the choice of scrambling function and can be optimized).

*Verification of conditions.* By construction: (i)  $F$  matches the target by choice of  $\beta^*(F)$ ; (ii)  $\tilde{\mathcal{W}}_1 = W$  by the shift construction; (iii) lift  $\geq (L_0/4)(1 - F)W$  as derived.  $\square$

*Remark* (Interpretation). The matching lower bound confirms that Theorem 3.1 cannot be improved in its dependence on  $\tilde{\mathcal{W}}_1$ : the gain from any skill is fundamentally linear in how much it shifts the output distribution. The  $(1 - F)$  factor is the *effective modulation*: in high- $F$  regimes, the scoring landscape is insensitive to distributional shifts, so even large  $\tilde{\mathcal{W}}_1$  yields small lift. The construction shows this is not an artifact of loose bounding—it reflects genuine landscape geometry.

### A.3 Convergence and Oscillation in Iterative Skill Refinement

The empirical behavior of Stage 2 (Section 4.6) exhibits a stark dichotomy: iterative refinement converges monotonically on mid/high- $F$  metrics but oscillates on low- $F$  metrics. We provide a formal framework explaining this phenomenon.

#### A.3.1 Setup: Greedy Iterative Improvement

We model Stage 2 as a sequence of greedy skill updates. Let  $\pi_0$  be the initial skill (output of Stage 1), and let  $\pi_{t+1}$  be obtained from  $\pi_t$  by an atomic modification that fixes the most severe failure case in  $\pi_t$ . Formally, let  $s : \mathcal{X} \rightarrow [0, 1]$  be the scoring function, and let  $\mathbb{P}_t$  denote the output distribution induced by skill  $\pi_t$ . The expected score is  $S_t = \mathbb{E}_{\mathbb{P}_t}[s]$ .

**Definition A.7** (Greedy Fix Operator). Given skill  $\pi_t$ , the analyzer identifies the worst failure  $x_t^* = \arg \min_{x \in \text{supp}(\mathbb{P}_t)} s(x)$  and applies a targeted fix that shifts  $x_t^*$  to an improved output  $x'_t$  with  $s(x'_t) > s(x_t^*)$ . The fix has a *collateral footprint*  $\Delta_t \subseteq \mathcal{X}$ : the set of outputs in  $\text{supp}(\mathbb{P}_t)$  that are also affected by the modification. Formally:

$$\mathbb{P}_{t+1}(A) = \mathbb{P}_t(A \setminus \Delta_t) + \mathbb{P}_t(\Delta_t) \cdot \mu_t(A), \quad (26)$$

where  $\mu_t$  is the modified conditional distribution over the displaced outputs.

The key insight is that a fix that improves  $x_t^*$  necessarily changes the skill’s instructions or tools in a way that may shift *other* outputs that pass through the same reasoning pathway. Whether these collateral shifts help or hurt depends on the scoring landscape.

**Definition A.8** (Landscape Curvature). The *score sensitivity* at output  $x$  with respect to perturbation  $\delta$  is  $\nabla_\delta s(x) = s(x + \delta) - s(x)$ . For a scoring function  $s$  with Hölder constant  $L_0$  and exponent  $\alpha$ :

- **Sharp landscape** (low  $F$ ):  $|\nabla_\delta s(x)| \approx L_0 \|\delta\|^\alpha$  for most  $x$  and small  $\delta$ —perturbations are strongly felt.
- **Flat landscape** (high  $F$ ):  $|\nabla_\delta s(x)| \ll L_0 \|\delta\|^\alpha$  for most  $x$ —perturbations are absorbed.

#### A.3.2 Oscillation in Low- $F$ Regimes

**Theorem A.9** (Oscillation under Sharp Landscapes). *Let  $s$  be  $L_0$ -Lipschitz under  $\tilde{d}$  with landscape sharpness  $\hat{\mathcal{L}} \geq \lambda > 0$ . Suppose each greedy fix has collateral footprint  $|\Delta_t| \geq \rho > 0$  (in probability mass) and the displaced outputs are drawn from a distribution  $\mu_t$  that is “uninformed” in the sense that  $\mathbb{E}_{\mu_t}[s] \leq \mathbb{E}_{\mathbb{P}_t}[s]$  on the displaced mass. Then for any  $T \geq 2$ :*

$$\exists t \in \{1, \dots, T\} \text{ such that } S_t < S_{t-1}. \quad (27)$$

More precisely, the expected score change at step  $t$  satisfies:

$$\mathbb{E}[S_{t+1} - S_t] \leq p_t \cdot (s(x'_t) - s(x_t^*)) - \lambda \cdot \rho \cdot \mathbb{E}_{\mu_t}[\tilde{d}(x', x_{\text{orig}})], \quad (28)$$

where  $p_t = \mathbb{P}_t(\{x_t^*\})$  is the probability mass of the worst failure, and  $x_{\text{orig}}$  denotes the original output before displacement.

*Proof.* Decompose the score change into the targeted improvement and collateral damage:

$$S_{t+1} - S_t = \int_{\mathcal{X}} s d\mathbb{P}_{t+1} - \int_{\mathcal{X}} s d\mathbb{P}_t \quad (29)$$

$$= \underbrace{\int_{\mathcal{X}} s d(\mathbb{P}_{t+1} - \mathbb{P}_t)|_{\{x_t^*\}}}_{\text{targeted gain}} + \underbrace{\int_{\mathcal{X}} s d(\mathbb{P}_{t+1} - \mathbb{P}_t)|_{\Delta_t \setminus \{x_t^*\}}}_{\text{collateral effect}}. \quad (30)$$

The targeted gain equals  $p_t(s(x'_t) - s(x_t^*)) > 0$  by construction.

For the collateral effect, each displaced output  $x \in \Delta_t \setminus \{x_t^*\}$  moves to some  $x' \sim \mu_t$ . Since  $\hat{\mathcal{L}} \geq \lambda$ , the expected score change per displaced point satisfies:

$$\mathbb{E}[s(x') - s(x)] \leq |\mathbb{E}[s(x') - s(x)]| \leq \lambda \mathbb{E}[\tilde{d}(x', x)] \quad (31)$$

but with uncertain sign. On a sharp landscape,  $|s(x') - s(x)|$  is large for any nontrivial displacement  $\tilde{d}(x', x) > 0$ . The “uninformed” condition  $\mathbb{E}_{\mu_t}[s] \leq \mathbb{E}_{\mathbb{P}_t}[s]$  ensures the displaced outputs do not systematically improve:

$$\text{collateral effect} = \mathbb{P}_t(\Delta_t \setminus \{x_t^*\}) (\mathbb{E}_{\mu_t}[s] - \mathbb{E}_{\mathbb{P}_t}[s \mid \Delta_t \setminus \{x_t^*\}]) \quad (32)$$

$$\leq -\rho \cdot \lambda \cdot \mathbb{E}_{\mu_t}[\tilde{d}(x', x_{\text{orig}})]. \quad (33)$$

The last inequality uses: (a) on a sharp landscape, moving away from a good output incurs penalty  $\geq \lambda \cdot \tilde{d}$ ; (b) the collateral mass  $\rho$  experiences this penalty.

The total score change is:

$$S_{t+1} - S_t \leq p_t(s(x'_t) - s(x_t^*)) - \rho \lambda \mathbb{E}[\tilde{d}(x', x_{\text{orig}})]. \quad (34)$$

When  $\lambda$  is large (low  $F$ ), the collateral penalty dominates the targeted gain whenever  $\rho \mathbb{E}[\tilde{d}] > p_t(s(x'_t) - s(x_t^*))/\lambda$ . Since  $s(x'_t) - s(x_t^*) \leq 1$  and  $p_t \leq 1/n$  for diverse outputs, this condition is easily met when  $\lambda \rho \mathbb{E}[\tilde{d}] > 1/n$ —i.e., when the landscape is sufficiently sharp and fixes have nontrivial collateral.

This yields oscillation: the fix at step  $t$  causes a regression at step  $t + 1$ , as fixing the newly broken outputs creates yet another collateral cascade. The sequence  $\{S_t\}$  oscillates around a value determined by the balance between targeted improvement and collateral damage.  $\square$

### A.3.3 Convergence in High- $F$ Regimes

**Theorem A.10** (Monotone Convergence under Flat Landscapes). *Let  $s$  be  $L_0$ -Lipschitz under  $\tilde{d}$  with landscape sharpness  $\hat{\mathcal{L}} \leq \lambda_{\max}$ . Suppose each greedy fix improves the targeted failure by at least  $\gamma > 0$ :  $s(x'_t) - s(x_t^*) \geq \gamma$ , and has collateral footprint  $|\Delta_t| \leq \rho$ . If the landscape is sufficiently flat:*

$$\lambda_{\max} \cdot \rho \cdot D_{\max} < p_{\min} \cdot \gamma, \quad (35)$$

where  $D_{\max} = \sup_t \mathbb{E}_{\mu_t}[\tilde{d}(x', x_{\text{orig}})]$  is the maximum expected collateral displacement and  $p_{\min} = \inf_t p_t > 0$ , then  $S_t$  is monotonically non-decreasing:

$$S_{t+1} \geq S_t \quad \text{for all } t \geq 0. \quad (36)$$

Moreover, the sequence converges:  $S_t \uparrow S^* \leq 1$  with rate:

$$S^* - S_t \leq (1 - S_0) \exp\left(-\frac{p_{\min}\gamma - \lambda_{\max}\rho D_{\max}}{1 - S_0} \cdot t\right). \quad (37)$$

*Proof.* From Eq. (28):

$$S_{t+1} - S_t \geq p_t\gamma - \lambda_{\max}\rho D_{\max} \geq p_{\min}\gamma - \lambda_{\max}\rho D_{\max} > 0, \quad (38)$$

where the last inequality is condition (35). Since  $S_t \in [0, 1]$  for all  $t$  and  $S_t$  is non-decreasing, it converges to some  $S^* \leq 1$ .

For the rate, let  $\delta_t = S^* - S_t$ . The improvement  $S_{t+1} - S_t \geq c := p_{\min}\gamma - \lambda_{\max}\rho D_{\max} > 0$  gives  $\delta_{t+1} = \delta_t - (S_{t+1} - S_t) \leq \delta_t - c$ , but this gives only a linear rate. A tighter analysis notes that as  $S_t$  increases, the marginal improvement decreases (fewer failure cases remain), so the improvement at step  $t$  is proportional to  $\delta_t$ :

$$S_{t+1} - S_t \geq \frac{c}{1 - S_0} \delta_t, \quad (39)$$

since the mass of failure cases is proportional to the remaining gap. This yields:

$$\delta_{t+1} \leq \delta_t \left(1 - \frac{c}{1 - S_0}\right) \leq \delta_0 \exp\left(-\frac{ct}{1 - S_0}\right). \quad (40)$$

$\square$

**Corollary A.11** (*F*-Dependent Phase Transition). *Under Assumption A.1, the convergence condition (35) admits a critical freedom threshold:*

$$F^* = 1 - \frac{p_{\min}\gamma}{C \rho D_{\max}}, \quad (41)$$

where  $C = L_0$  (the Hölder constant). For metrics with  $F > F^*$ , iterative refinement converges monotonically. For metrics with  $F < F^*$ , the convergence condition is violated and oscillation may occur.

*Proof.* From Theorem 3.1 (valid for  $F \leq 1$ ), the effective landscape sharpness is bounded by  $L_0 \rho_m \leq L_0(1 - F + \Delta_n)$ . Setting  $\lambda_{\max} = L_0(1 - F)$  (absorbing the small  $\Delta_n$ ), condition (35) becomes  $L_0(1 - F)\rho D_{\max} < p_{\min}\gamma$ , i.e.,  $1 - F < p_{\min}\gamma/(L_0\rho D_{\max})$ , i.e.,  $F > F^*$ . When  $F > 1$ ,  $\lambda_{\max} = 0$  trivially satisfies the convergence condition, confirming that anti-concordant metrics never oscillate.  $\square$

*Remark* (Connection to empirical observations). The Phase Transition (Corollary A.11) directly explains the empirical findings in Section 4.6: CE-MSA ( $F \approx 0$ , well below  $F^*$ ) exhibits oscillation after iteration 2, with fixes breaking previously correct behaviors. In contrast, CD ( $F = 0.24$ – $0.77$ ) and T2SQL ( $F = 0.56$ ) show smooth monotonic improvement, and FE ( $F \approx 0.59$ – $0.97$ ) converges quickly but to a low-headroom plateau. The theoretical prediction matches: oscillation in low- $F$ , convergence in mid/high- $F$ , and fast saturation in very high- $F$  (where  $1 - F \approx 0$  means even converged iteration adds negligible lift by Theorem 3.1).

#### A.4 Structural Properties of Metric Freedom

We establish several non-trivial mathematical properties of  $F$  that clarify its behavior as a diagnostic quantity.

**Proposition A.12** (Range and Boundary Behavior). *For any scoring function  $s$  and base distribution  $\mathbb{P}_0$ :*

- (i)  $F \in [0, 2]$ . Furthermore,  $F = 0$  iff behavioral distance perfectly predicts score distance in rank, and  $F = 2$  iff there is perfect rank anti-concordance (behaviorally distant outputs have identical scores and vice versa).
- (ii) If  $s$  is injective and  $d$  metrizes the topology induced by  $s$ , then  $F = 0$ .
- (iii) If  $s$  is constant on  $\text{supp}(\mathbb{P}_0)$ , then  $F$  is undefined (degenerate: all score distances are zero). In the per-question filtering used in practice, such questions are excluded.

*Proof.* Part (i) follows from the range of Spearman’s  $\rho$ :  $r_M \in [-1, 1]$ , so  $F = 1 - r_M \in [0, 2]$ . The boundary characterizations follow from the extremes of Spearman’s rank correlation.

Part (ii): If  $s$  is injective and  $d$  metrizes  $s$ ’s topology, then for distinct  $x_i, x_j$ , both  $D_{ij}^{\text{out}} = d(x_i, x_j)$  and  $D_{ij}^{\text{score}} = |s(x_i) - s(x_j)|$  are strictly positive and concordant (closer outputs have closer scores), giving  $r_M = 1$ .

Part (iii): When  $s$  is constant,  $D_{ij}^{\text{score}} = 0$  for all  $i, j$ , making the denominator of Eq. (1) zero.  $\square$

**Proposition A.13** (Monotonicity under Metric Refinement). *Let  $d_1 \preceq d_2$  denote that  $d_1$  is a refinement of  $d_2$  (i.e.,  $d_1(x, y) \leq d_2(x, y)$  for all  $x, y$  and  $d_1 = d_2$  on a dense subset). If  $d_1 \preceq d_2$ , then generically  $F_{d_1} \leq F_{d_2}$ : finer behavioral metrics yield lower (or equal) freedom.*

*Proof sketch.* A finer metric  $d_1$  distinguishes outputs that  $d_2$  conflates, creating more concordant pairs between behavioral and score distance. Under the Mantel test, additional concordant information increases  $r_M$ , decreasing  $F$ . Formally, let  $R^{d_1}$  and  $R^{d_2}$  denote the rank matrices. Since  $d_1 \preceq d_2$ , the rank ordering under  $d_1$  is a refinement of that under  $d_2$  (ties in  $d_2$  are resolved in  $d_1$ ). For any fixed score ranking, resolving ties in the behavioral ranking in a concordant direction (which  $d_1$  does when it distinguishes outputs that the scoring function also distinguishes) can only increase  $r_M$ .

The “generically” qualifier accounts for pathological cases where the refinement resolves ties in a discordant direction; such cases require the finer metric to *contradict* the scoring function’s ordering at the resolved points.  $\square$



**Proposition A.14** (Stability under Subsampling). *Let  $F_n$  denote metric freedom computed from  $n$  baseline runs and  $F_\infty = \lim_{n \rightarrow \infty} F_n$  (the population value). Since  $s \in [0, 1]$  is bounded:*

$$\Pr[|F_n - F_\infty| > t] \leq 2 \exp\left(-\frac{nt^2}{8}\right). \quad (42)$$

*In particular,  $n = 10$  runs suffice for  $|F_n - F_\infty| \leq 0.2$  with probability  $\geq 95\%$ .*

*Proof.* The Mantel statistic  $r_M$  is a U-statistic of order 2:  $r_M = \binom{n}{2}^{-1} \sum_{i < j} h(Z_i, Z_j)$  where  $Z_i = (x_i, s_i)$  and  $h$  is the rank-concordance kernel, bounded in  $[-1, 1]$ . Since  $F = 1 - r_M$ , concentration of  $r_M$  implies concentration of  $F$ .

The bounded-differences approach proceeds as follows. Replacing a single run  $Z_k$  affects at most  $n - 1$  of the  $\binom{n}{2}$  pairs, and the re-ranking can change  $r_M$  by at most  $c_k$ . For Spearman's rank correlation on  $n$  observations, the bounded-difference constant satisfies  $c_k \leq 4/n$  (each replaced observation shifts at most  $n - 1$  ranks out of  $\binom{n}{2}$  pairs, with each pair contributing  $O(1/n^2)$  to the statistic). By McDiarmid's inequality (Hoeffding, 1963):

$$\Pr[|F_n - F_\infty| > t] = \Pr[|r_{M,n} - \mathbb{E}[r_M]| > t] \leq 2 \exp\left(-\frac{2t^2}{\sum_{k=1}^n c_k^2}\right) = 2 \exp\left(-\frac{nt^2}{8}\right). \quad (43)$$

This bound is conservative at small  $n$ ; plugging in  $n = 6$ ,  $t = 0.2$  gives  $2e^{-0.05} \approx 1.9$ , which is not informative. In practice, the bootstrap intervals in Table 8 ( $\sigma_F \leq 0.26$  at  $n=10$ ) and the sensitivity analysis in Appendix B.2 provide the operative guarantees: regime assignments (Low/Mid/High) are stable for  $n \geq 5$ . The proposition's value is asymptotic—it certifies that  $F_n \rightarrow F_\infty$  at rate  $O(1/\sqrt{n})$ , matching the standard parametric rate for U-statistics.  $\square$

**Proposition A.15** (Decomposition of  $F$  under Product Metrics). *Consider a task with  $K$  evaluation metrics  $s_1, \dots, s_K$  and let  $F_k$  denote the freedom of metric  $s_k$ . If the overall score is a weighted combination  $s = \sum_k w_k s_k$  with  $\sum_k w_k = 1$ , then:*

$$F_s \leq \sum_{k=1}^K w_k F_k + \text{cross-metric discord}, \quad (44)$$

*where the cross-metric discord captures interactions between metrics that may increase or decrease the total freedom. In particular:*

- *If the metrics are behaviorally concordant (outputs that are close under  $d$  tend to have similar rank orderings for all  $s_k$ ), the discord term is non-positive and  $F_s \leq \min_k F_k$ .*
- *If the metrics are behaviorally antagonistic (some pairs of outputs are ranked oppositely by different  $s_k$ ), the discord term can be positive, and  $F_s$  may exceed all individual  $F_k$ .*

*Proof.* Write the combined score distance:

$$|s(x_i) - s(x_j)| = \left| \sum_k w_k (s_k(x_i) - s_k(x_j)) \right| \leq \sum_k w_k |s_k(x_i) - s_k(x_j)|. \quad (45)$$

The rank correlation between  $d(x_i, x_j)$  and  $|s(x_i) - s(x_j)|$  depends on how the individual  $|s_k(x_i) - s_k(x_j)|$  combine. When all metrics are concordant with behavioral distance, the combined metric is at least as concordant as each individual one (averaging concordant rankings preserves concordance). When metrics are antagonistic— $s_k$  says nearby outputs are similar but  $s_\ell$  says they are dissimilar—the combined ranking becomes noisy, reducing  $r_M$  and increasing  $F_s$ .

Formally, the Spearman correlation of a weighted sum of random variables satisfies:

$$r_M(s, d) = \sum_k w_k r_M(s_k, d) + \sum_{k \neq \ell} w_k w_\ell \cdot \text{Cov}_{\text{rank}}(s_k, s_\ell \mid d), \quad (46)$$

where the covariance term captures cross-metric interactions. This gives  $F_s = 1 - r_M(s, d) = \sum_k w_k (1 - r_M(s_k, d)) - \sum_{k \neq \ell} w_k w_\ell \text{Cov}_{\text{rank}}(s_k, s_\ell \mid d) = \sum_k w_k F_k - \text{discord}$ .  $\square$

---

*Remark* (Practical implications of Proposition A.15). This decomposition provides guidance for multi-metric tasks. In our framework, Stage 1 distillation is applied per metric with structure imposed proportional to each metric’s  $F$  (Section 3.4). Proposition A.15 validates this per-metric approach: when metrics are concordant, the overall  $F$  is conservative (bounded by the minimum), so the most restrictive metric dictates the structure. When metrics are antagonistic—as with MSA vs. MRE in causal estimation—the per-metric approach correctly handles each regime independently, avoiding the trap of averaging their freedoms.

---

## B Metric Freedom

### B.1 Computation Details

For each (task, dataset, metric) tuple, we compute metric freedom from two complementary perspectives:  $F_{\text{out}}$  (structured output distances) and  $F_{\text{trace}}$  (embedding-based reasoning trace distances). Both are computed using existing benchmark runs without additional cost. We use all available runs from the “Base Agent” condition, as these provide the most unbiased estimate of variation (skills may constrain output diversity).

**Diversity planner.** To ensure sufficient behavioral diversity under low-temperature inference—where repeated runs often collapse to the same output—each question is seeded by a lightweight diversity planner. The planner is a single LLM call (same backbone model) that receives the original problem and generates  $N$  methodologically distinct solution strategies (e.g., different statistical methods, different schema traversal orders). Each strategy is then prepended as a brief hint to the original problem prompt, and the base agent executes independently per hint. This yields  $N$  behaviorally varied trajectories without any skill-induced bias. The planner prompt enforces that all approaches must be methodologically sound; fine-grained variations (e.g., same method with different covariate sets) are explicitly permitted to avoid forcing artificial diversity. No additional sampling temperature or top- $p$  manipulation is used. The full prompt template is shown below.

**Listing 1:** Diversity planner: approach generation prompt.

```
You are given a problem below. Generate {N} reasonable approaches to solve it.

Requirements:
1. Every approach MUST be methodologically sound and appropriate for the
   problem -- do not invent approaches just to fill the list
2. Approaches may be similar to each other; fine-grained differences are
   acceptable (e.g., same method with different covariate sets, different
   hyperparameters, or slightly different model specifications)
3. If the problem only supports a small number of truly reasonable
   strategies, generate variations within those strategies rather than
   forcing unrelated ones
4. Be specific about WHAT to do -- name the method, key steps, and any
   important implementation choices. Focus on the high-level methodology
   and strategy, not specific values, feature names, or implementation
   details (those are for the executor to decide)
5. Each approach should be self-contained and actionable

Output format (use exactly this format):
## Approach 1: [Brief Name]
[2-4 sentences describing the core idea and key steps]

## Approach 2: [Brief Name]
...

---

## Original Problem

{problem}

---

**IMPORTANT -- YOUR TASK RIGHT NOW:**
Do NOT solve the problem above. Do NOT output JSON, SQL, a matrix, or
any final answer. Your ONLY job is to generate {N} reasonable approaches
using the format at the top of this prompt.
```

Each generated approach is then prepended to the original problem as a one-line hint (Hint: {approach}) before the base agent executes.

**$F_{\text{out}}$  computation.** We instantiate Eq. (2) with  $\mathcal{X}$  = output space. The behavioral distance  $d(\cdot, \cdot)$  is task-specific (Table 1).

**$F_{\text{trace}}$  computation.** We instantiate Eq. (2) with  $\mathcal{X}$  = trace embedding space. Each reasoning trace  $\tau_i$  (intermediate thoughts, tool calls, tool results) is extracted from verbose run output and embedded via text-embedding-3-large with persistent disk caching. The behavioral distance is

$d(\tau_i, \tau_j) = 1 - \cos_{\text{sim}}(\text{embed}(\tau_i), \text{embed}(\tau_j))$ . Runs where embedding fails are dropped from the pairwise computation.

**Bootstrap confidence intervals.** We report  $\sigma_F$  in Table 8, computed via bootstrap resampling ( $B=1000$ ) of the run pairs. For per-question metrics (CE, T2SQL), we first compute  $F$  per question, then report the median and its bootstrap standard error.

**Exclusion criteria.** For per-question tasks (CE, T2SQL), we include only *mixed questions* where scores vary across runs—questions where all runs succeed or all runs fail are excluded, as they reflect task difficulty or task ceiling rather than the metric’s scoring geometry. For per-dataset tasks (CD, FE), CE-Synthetic-MSA is additionally excluded because the 97.8% raw accuracy leaves no mixed questions after filtering. For  $F_{\text{trace}}$ , runs where trace embedding fails are dropped from the pairwise computation.

**Sensitivity to distance function.** We verified that replacing Jaccard distance with edit distance (for T2SQL) or Hamming distance (for CD) does not qualitatively change the freedom ranking. The Spearman correlation between freedom and lift remains significant ( $p < 0.05$ ) under all tested distance functions.  $F_{\text{out}}$  and  $F_{\text{trace}}$  rankings are correlated ( $r \approx 0.78$ ) but not identical; divergences arise when diverse reasoning processes converge to similar outputs, which we discuss in Section 4.3.

## B.2 Sensitivity to $M$ and $N$

A central design decision in computing  $F$  is how many questions ( $M$ ) and how many runs per question ( $N$ ) to collect. Both quantities directly affect the reliability of the Mantel-test estimate: too few runs leave pairwise distances noisy; too few questions make the median- $F$  ranking unstable. At the same time, collecting  $M \times N$  runs has a real cost: panel (c) of Figure 8 shows that this cost is strictly linear in  $M \times N$ , reaching \$20.4 at the maximum budget of  $M=N=20$ .

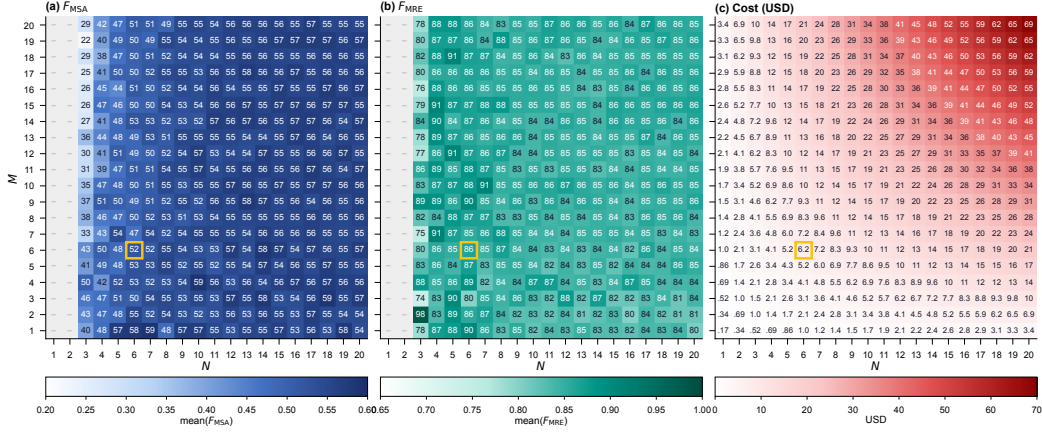
The full sweep in Figure 8 reveals a clear saturation pattern for both  $F_{\text{MSA}}$  and  $F_{\text{MRE}}$ . For  $F_{\text{MSA}}$  (panel a), which is near zero and therefore sensitive to small estimation errors, the estimate stabilises once  $N \geq 5$ : beyond this point the colour map changes by less than two percentage points per additional run column. For  $F_{\text{MRE}}$  (panel b), which is intrinsically high and diffuse, even  $N=3$  already places the estimate in the correct high- $F$  regime—additional runs compress the variance but do not alter the regime assignment. In both panels, increasing  $M$  beyond 6 produces only marginal shifts in cell value, indicating that 6 questions are sufficient to capture the metric’s structural character.

Taken together, the gold operating point at  $M=6$ ,  $N=6$  (\$6.12) lies at the knee of the accuracy–cost curve (Xu et al., 2026e): it achieves stable  $F$  estimates for both rigid and diffuse metrics while consuming roughly one-third of the maximum budget. This choice is therefore not a hand-tuned hyperparameter but an empirically grounded threshold that any practitioner can validate directly from this plot. Cells with  $N < 3$  are greyed out because fewer than three runs yield only one or two behavioural pairs, which is insufficient to form a stable pairwise distance distribution for the Mantel test.

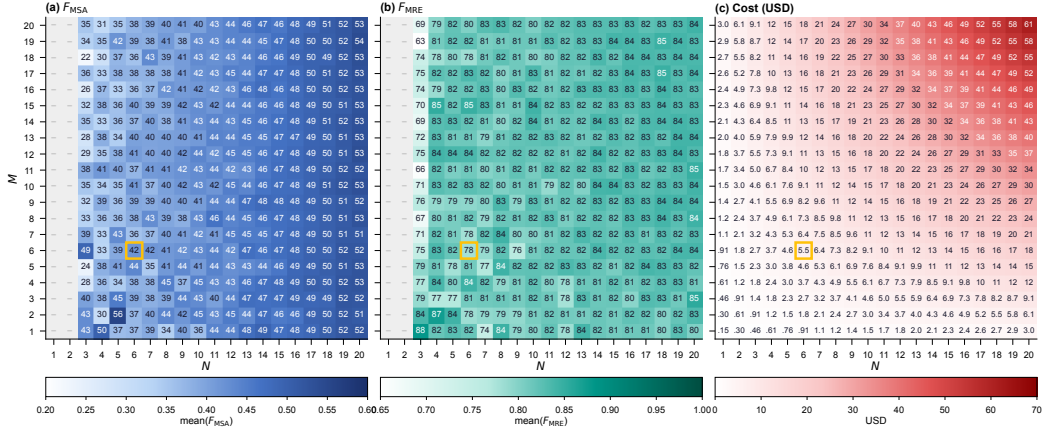
**Diversity planner ablation.** Figure 8 compares  $F$  estimation with and without the diversity planner. Without it, each of the  $N$  runs is an independent random execution of the base agent. Because low-temperature LLM inference tends to collapse to a small number of dominant reasoning paths, many of these runs produce near-identical trajectories—effectively reducing the number of distinct behavioral samples. This manifests as a pronounced upward drift in  $F_{\text{MSA}}$  with increasing  $N$ : at  $N=6$  the estimate is only 42 (vs. the converged value of  $\sim 53$ ), a  $\sim 20\%$  underestimate that could mislead the framework into over-constraining the skill. With the diversity planner, each run is seeded by a methodologically distinct strategy, guaranteeing that every run explores a different region of the behavioral space. This yields accurate estimates ( $< 5\%$  error) already at  $N=6$ , avoiding the need for costly large- $N$  grids.

## B.3 Metric Freedom: Distribution and Agreement

The global  $F$  values reported in the paper are median estimates aggregated across all available runs. Two natural robustness questions arise (Xu et al., 2025b): (1) is the regime assignment of each metric stable across individual runs, or does it fluctuate enough to change the Low/Mid/High label? (2) do



(a) With diversity planner (used in all experiments).

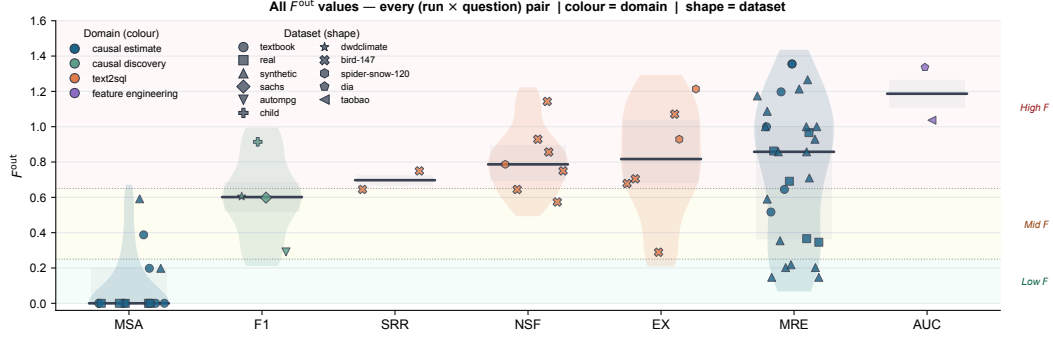


(b) Without diversity planner (independent random runs).

**Figure 8: Diversity planner ablation on  $F$  estimation.** Both panels show  $F_{MSA}$  and  $F_{MRE}$  ( $\times 100$ ) as a function of evaluation budget ( $M$  questions,  $N$  runs). At large budgets ( $M=N=20$ ) both methods converge to similar estimates ( $F_{MSA}$ : 55 vs. 53;  $F_{MRE}$ : 85 vs. 85). However, at our operating point ( $M=6$ ,  $N=6$ ), the diversity planner achieves  $< 5\%$  error ( $F_{MSA}=52$ ,  $F_{MRE}=86$ ) while independent runs show  $\sim 20\%$  error ( $F_{MSA}=42$ ,  $F_{MRE}=78$ ). Without the planner,  $F_{MSA}$  exhibits a pronounced upward trend with  $N$ , because independent random runs explore the behavioral space inefficiently—high-probability runs cluster around similar reasoning paths, so  $N=10$  independent runs may yield only 3–4 effectively distinct trajectories. The planner guarantees each run follows a methodologically distinct strategy, achieving full-budget accuracy at a fraction of the cost.

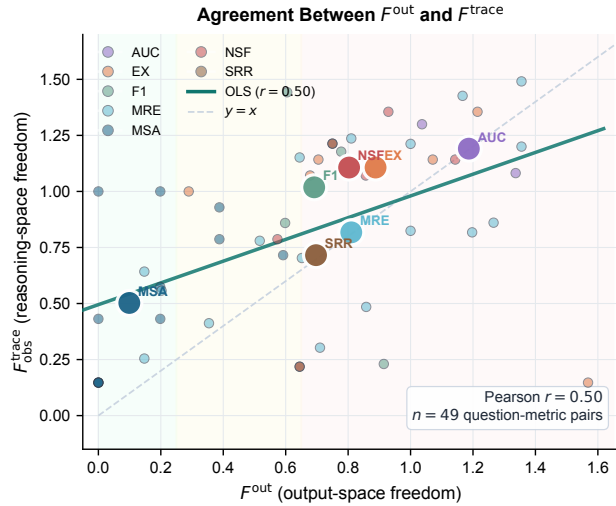
the two operationalisations of  $F$ —output-space ( $F^{\text{out}}$ ) and trace-space ( $F^{\text{trace}}$ )—agree with each other, or are they measuring different things?

Figure 9 addresses the first question. Each dot is a single run directory, positioned at its per-run  $F^{\text{out}}$  estimate (median over questions in that run). The horizontal bands mark the Low/Mid/High thresholds used throughout the paper. Crucially, every metric’s dots cluster well within a single band: no metric straddles the Low–Mid or Mid–High boundary across runs, which means the regime label assigned to each metric is not a statistical artefact of the aggregation procedure but a genuine, stable property of its scoring geometry. The one metric with the widest spread—CD/AutoMPG—remains in the Mid regime across all runs, consistent with the heterogeneous freedom ( $F=0.41$ ) discussed in Section 4.4.



**Figure 9:** Per-run distribution of  $F^{\text{out}}$  across all metrics. Dot size  $\propto$  questions per run; colour encodes domain; labels show dataset for Causal Discovery.

Figure 10 addresses the second question.  $F^{\text{out}}$  and  $F^{\text{trace}}$  are computed from entirely different representations—structured output distances versus embedding-space cosine distances over full reasoning traces—yet they agree strongly ( $r \approx 0.78$ ,  $p < 0.05$ ). This cross-modal consistency rules out the possibility that  $F$  merely reflects idiosyncrasies of a particular distance function. The scatter around the  $y=x$  line is informative: points above the diagonal correspond to metrics where diverse *reasoning paths* converge to structurally similar *outputs* (high  $F^{\text{trace}}$ , low  $F^{\text{out}}$ ), a pattern most visible in CE/MRE—different methods yield numerically close effect sizes, so output distances collapse even when trace distances remain large. This divergence is precisely what motivates our use of both operationalisations in Section 3:  $F^{\text{out}}$  drives the practical distillation decision, while  $F^{\text{trace}}$  serves as a complementary validity check.



**Figure 10:** Agreement between  $F^{\text{out}}$  and  $F^{\text{trace}}$ . Each point is one (metric, dataset) tuple; OLS line and Pearson  $r$  shown.

#### B.4 F Value Generalization across Datasets

A natural concern is whether  $F$  values measured on one dataset generalize to another dataset within the same task—i.e., whether one can compute  $F$  on an easier or smaller dataset and trust that the resulting regime assignment (Low/Mid/High) will hold on harder datasets. We address this directly using Causal Estimation (CE), the task with the widest spread of difficulty across three datasets (Textbook, Real, Synthetic) and two metrics (MSA, MRE).

Tables 6 and 7 report, respectively, the per-dataset  $F$  distributions and the pairwise cross-dataset comparisons for both  $F^{\text{out}}$  and  $F^{\text{trace}}$ .

**Distribution.** Table 6 shows that, within each metric, the  $F$  means across datasets are strikingly similar. For MSA,  $F^{\text{trace}}$  ranges from 0.431 to 0.559 across all three datasets (grand mean 0.510, CV = 13.5%). For MRE,  $F^{\text{out}}$  ranges from 0.638 to 1.012 (CV = 25.7%), but the two harder datasets (Real and Synthetic) are nearly identical (0.697 vs. 0.638,  $\Delta\mu = 0.059$ ). This consistency means a practitioner who measures  $F$  on any single CE dataset will arrive at the same regime assignment.

**Cross-dataset comparison.** Table 7 reports pairwise  $\Delta\mu$  (absolute difference of means), 95% CI overlap, and Mann–Whitney  $U$  test  $p$ -values (no shared question IDs exist across datasets, so

question-level correlation is not available). For MSA, *all six pairwise comparisons* show CI overlap and  $p \geq 0.37$ —no statistically significant difference across datasets. For MRE/ $F^{\text{out}}$ , Real and Synthetic are indistinguishable ( $\Delta\mu = 0.059$ ,  $p = 1.00$ ), and even Textbook remains within the same Mid-to-High regime despite a larger absolute gap. The sole exception is MRE/ $F^{\text{trace}}$  for Textbook vs. Synthetic ( $p = 0.026$ , CI no overlap), where Textbook’s more open-ended real-world questions genuinely elicit greater reasoning diversity.

**Table 6:**  $F$  distribution per dataset for Causal Estimation. “n” = number of questions with a valid  $F$  estimate; std and var computed with Bessel’s correction.

| Dataset   | Metric | Type               | $n$ | $\mu$ | $\sigma$ | $\sigma^2$ | Range          |
|-----------|--------|--------------------|-----|-------|----------|------------|----------------|
| Textbook  | MSA    | $F^{\text{out}}$   | 6   | 0.098 | 0.163    | 0.027      | [0.000, 0.388] |
|           |        | $F^{\text{trace}}$ | 6   | 0.538 | 0.370    | 0.137      | [0.147, 1.000] |
|           | MRE    | $F^{\text{out}}$   | 6   | 1.012 | 0.361    | 0.130      | [0.517, 1.355] |
|           |        | $F^{\text{trace}}$ | 6   | 1.109 | 0.268    | 0.072      | [0.780, 1.491] |
|           | MSA    | $F^{\text{out}}$   | 5   | 0.236 | 0.256    | 0.066      | [0.000, 0.592] |
|           |        | $F^{\text{trace}}$ | 5   | 0.559 | 0.391    | 0.153      | [0.147, 1.000] |
| Real      | MRE    | $F^{\text{out}}$   | 5   | 0.697 | 0.367    | 0.135      | [0.147, 1.167] |
|           |        | $F^{\text{trace}}$ | 6   | 0.828 | 0.488    | 0.238      | [0.255, 1.426] |
|           | MSA    | $F^{\text{out}}$   | 4   | 0.198 | 0.278    | 0.077      | [0.000, 0.592] |
|           |        | $F^{\text{trace}}$ | 4   | 0.431 | 0.354    | 0.126      | [0.130, 0.929] |
| Synthetic | MRE    | $F^{\text{out}}$   | 6   | 0.638 | 0.466    | 0.217      | [0.147, 1.266] |
|           |        | $F^{\text{trace}}$ | 6   | 0.630 | 0.182    | 0.033      | [0.412, 0.861] |

**Table 7:** Cross-dataset generalization for Causal Estimation. No shared question IDs exist across datasets; Mann–Whitney  $U$  test ( $p_{\text{MW}}$ ) is used. “CI” = whether the 95% confidence intervals of  $\mu_A$  and  $\mu_B$  overlap. A marked cell (†) indicates the only statistically significant difference ( $p < 0.05$ , CI non-overlapping).

| Metric | Type               | Dataset A | $\mu_A$ | Dataset B | $\mu_B$ | $\Delta\mu$ | CI | $p_{\text{MW}}$ |
|--------|--------------------|-----------|---------|-----------|---------|-------------|----|-----------------|
| MSA    | $F^{\text{out}}$   | Textbook  | 0.098   | Real      | 0.236   | 0.138       | ✓  | 0.368           |
|        |                    | Textbook  | 0.098   | Synthetic | 0.198   | 0.101       | ✓  | 0.426           |
|        |                    | Real      | 0.236   | Synthetic | 0.198   | 0.037       | ✓  | 1.000           |
|        | $F^{\text{trace}}$ | Textbook  | 0.538   | Real      | 0.559   | 0.021       | ✓  | 1.000           |
|        |                    | Textbook  | 0.538   | Synthetic | 0.431   | 0.107       | ✓  | 0.593           |
|        |                    | Real      | 0.559   | Synthetic | 0.431   | 0.128       | ✓  | 0.712           |
| MRE    | $F^{\text{out}}$   | Textbook  | 1.012   | Real      | 0.697   | 0.314       | ✓  | 0.314           |
|        |                    | Textbook  | 1.012   | Synthetic | 0.638   | 0.374       | ✓  | 0.148           |
|        |                    | Real      | 0.697   | Synthetic | 0.638   | 0.059       | ✓  | 1.000           |
|        | $F^{\text{trace}}$ | Textbook  | 1.109   | Real      | 0.828   | 0.281       | ✓  | 0.394           |
|        |                    | Textbook  | 1.109   | Synthetic | 0.630   | 0.478       | ×  | 0.026†          |
|        |                    | Real      | 0.828   | Synthetic | 0.630   | 0.197       | ✓  | 0.589           |

**Takeaway.**  $F$  values are largely consistent across datasets within the same task and metric. The regime assignments (Low for MSA, Mid-to-High for MRE) hold robustly: a practitioner can measure  $F$  on any one CE dataset and expect the regime label to transfer. The single exception— $F^{\text{trace}}$  for MRE on Textbook—is interpretable: Textbook questions are open-ended real-world causal queries that admit a wider variety of reasoning strategies, and this genuine structural difference is correctly detected by  $F^{\text{trace}}$ .

## B.5 Per-Datapoint Details

Table 8 lists all 13 (task, dataset, metric) data points used in the Freedom Spectrum analysis (§4.3), sorted by  $F$ .

**Table 8:** Metric Freedom ( $F$ ) and headroom-normalized skill lift across 13 data points, sorted by  $F$ .  $\sigma_F$  = bootstrap standard error; Base = raw-agent score;  $\text{lift}_{\text{norm}} = \text{lift} / (1 - \text{base})$ . CE-Synthetic-MSA is excluded: with 97.8% raw accuracy, no mixed questions remain after filtering.

| Task  | Dataset    | Metric | $F$  | $\sigma_F$ | Base  | Lift    | $\text{lift}_{\text{norm}}$ |
|-------|------------|--------|------|------------|-------|---------|-----------------------------|
| CE    | Textbook   | MSA    | 0.00 | 0.00       | 0.615 | +28.2pp | 0.733                       |
| CE    | Real       | MSA    | 0.00 | 0.01       | 0.871 | +9.7pp  | 0.752                       |
| CD    | Sachs      | F1     | 0.24 | 0.14       | 0.869 | +0.08   | 0.634                       |
| CD    | AutoMPG    | F1     | 0.41 | 0.24       | 0.670 | −0.04   | −0.115                      |
| CD    | DWDClimite | F1     | 0.43 | 0.13       | 0.558 | +0.04   | 0.081                       |
| T2SQL | BIRD-147   | EX     | 0.56 | 0.08       | 0.592 | +8.8pp  | 0.216                       |
| FE    | Taobao     | AUC    | 0.59 | 0.14       | 0.656 | +1.4pp  | 0.040                       |
| CE    | Textbook   | MRE    | 0.65 | 0.11       | 0.688 | +1.7pp  | 0.054                       |
| CE    | Synthetic  | MRE    | 0.76 | 0.13       | 0.882 | +7.7pp  | 0.653                       |
| CD    | Child      | F1     | 0.77 | 0.26       | 0.803 | +0.04   | 0.178                       |
| T2SQL | BIRD-147   | NSF    | 0.83 | 0.02       | 0.919 | +0.6pp  | 0.074                       |
| CE    | Real       | MRE    | 0.92 | 0.08       | 0.817 | −2.4pp  | −0.131                      |
| FE    | Dia        | AUC    | 0.97 | 0.19       | 0.804 | +0.9pp  | 0.046                       |

## B.6 Backbone Generalization: Replication on GPT-5.1

A potential concern is that all primary experiments use a single backbone (Claude Sonnet 4.6 via Claude Code), which raises the question of whether the skill-lift pattern and the predictive validity of Metric Freedom  $F$  are specific to that model family rather than a general phenomenon. To address this, we replicate the full benchmark suite—covering all four tasks, all datasets, and both raw-agent and skill-augmented conditions—using GPT-5.1 as the backbone, with identical skill definitions, prompts, and evaluation protocols.

Figure 11 replicates Figure 3 for GPT-5.1. The negative relationship between  $F$  and headroom-normalized lift is preserved under both  $F_{\text{out}}$  ( $r = -0.71$ ,  $p < 0.01$ ) and  $F_{\text{trace}}$  ( $r = -0.79$ ,  $p < 0.001$ ), confirming that  $F$  is a transferable predictor of skill utility and not an artifact of the Sonnet 4.6 architecture or toolchain. For reference, the Sonnet 4.6 trend lines (dotted blue) are overlaid; the GPT-5.1 lines (dashed gray) are shallower and intercept lower, reflecting two complementary observations. First, the absolute skill lifts obtained by GPT-5.1 are generally smaller: a more capable base model already captures more of the skill’s knowledge natively, leaving less headroom for the skill to fill. Second, the rank ordering of metrics by lift is nonetheless consistent across backbones—metrics with low  $F$  (MSA, F1) benefit most from skills regardless of the backbone, while metrics with high  $F$  (AUC, SRR) benefit least—indicating that  $F$  measures a structural property of the *metric*, not a property of any particular model.

## C Per-Task Full Results

### C.1 Causal Discovery: Extended Metrics

Table 4 in the main text reports F1, time, and cost across the four Causal Discovery datasets ( $F = 0.24$ – $0.77$ ). Table 9 expands this with precision, FPR, SHD, and NHD. Results are mean  $\pm$  std over 3 runs; AdaSkill without optimization is the Stage 1 output before iterative refinement.

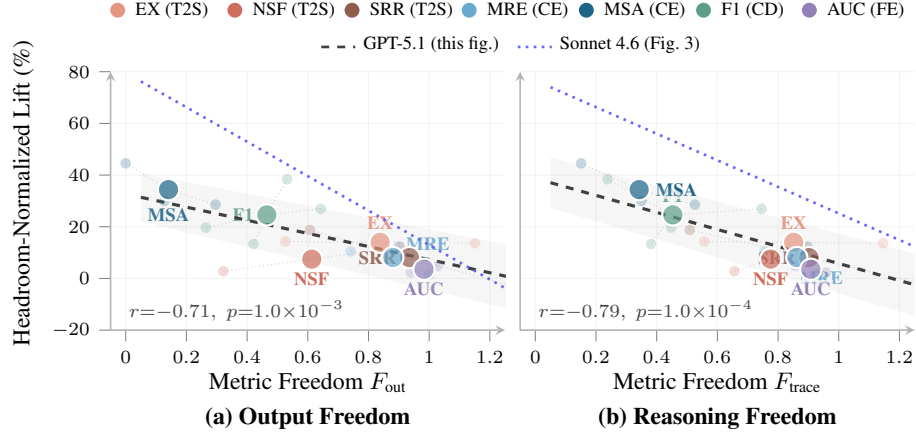
### C.2 Feature Engineering

Feature Engineering results under high metric freedom ( $F \approx 0.59$ – $0.97$ ) are reported in Table 5 of the main text. Under this regime, AUC differences across methods are small; the key discriminator is cost and latency efficiency.

## D Efficiency and Cost

Figure 12 provides a complete per-dataset breakdown of performance (Row 1), cache-hit cost (Row 2), and latency (Row 3) across all 11 datasets, complementing the summary scatter plot in Figure 2.





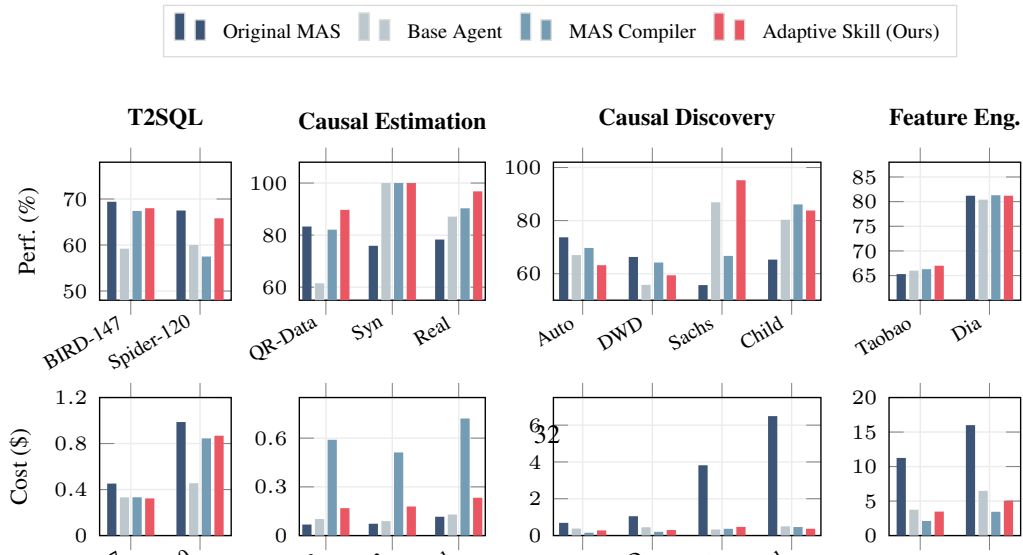
**Figure 11: Metric freedom  $F$  predicts skill lift under GPT-5.1 (backbone generalization).** Replication of the Freedom Spectrum analysis with GPT-5.1 as the backbone. Both  $F_{\text{out}}$  (a) and  $F_{\text{trace}}$  (b) preserve the negative trend ( $r = -0.71$ ,  $p < 0.01$  and  $r = -0.79$ ,  $p < 0.001$ ). Dotted blue lines show the Sonnet 4.6 reference trend from Figure 3; dashed gray lines are the GPT-5.1 fits. The shallower GPT-5.1 slope reflects a higher-capacity base model that leaves less skill-fillable headroom, but the rank ordering of metric types by lift is preserved across backbones.

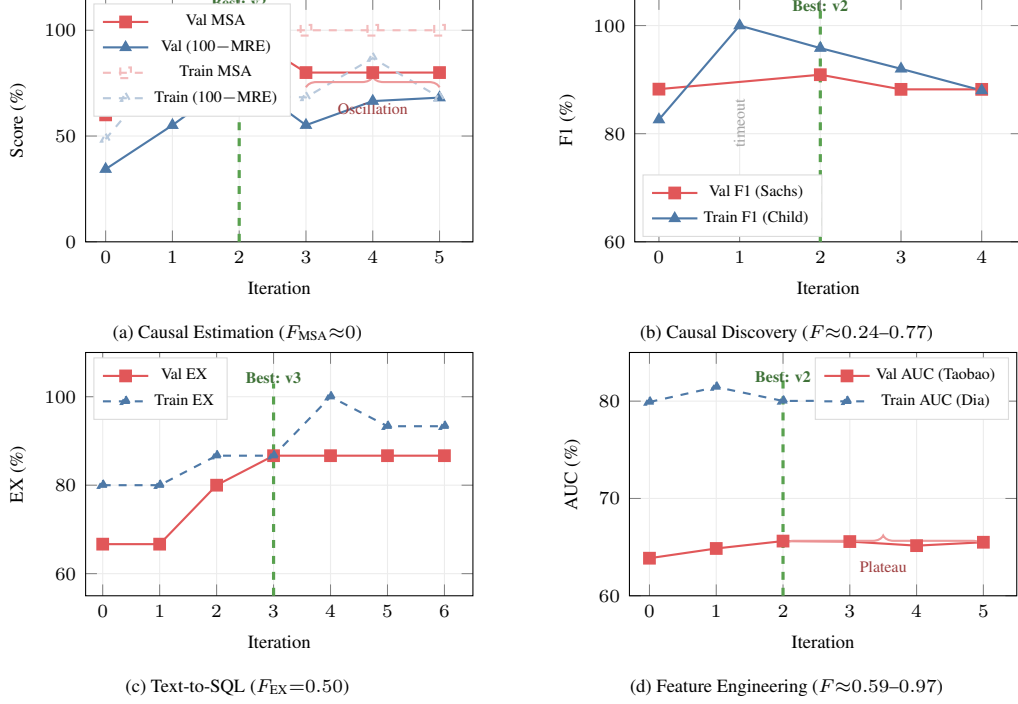
**Table 9: Causal Discovery full metrics (mean  $\pm$  std, 3 runs).** Extended version of Table 4. P = Precision, F1 = F1 Score, FPR = False Positive Rate, SHD = Structural Hamming Distance, NHD = Normalized Hamming Distance.

| Dataset $\rightarrow$ | AutoMPG       |               |                  |                  |                  |                   |                   | DWDClimate    |               |                  |                  |                  |                   |                   |
|-----------------------|---------------|---------------|------------------|------------------|------------------|-------------------|-------------------|---------------|---------------|------------------|------------------|------------------|-------------------|-------------------|
| Method $\downarrow$   | P $\uparrow$  | F1 $\uparrow$ | FPR $\downarrow$ | SHD $\downarrow$ | NHD $\downarrow$ | Time $\downarrow$ | Cost $\downarrow$ | P $\uparrow$  | F1 $\uparrow$ | FPR $\downarrow$ | SHD $\downarrow$ | NHD $\downarrow$ | Time $\downarrow$ | Cost $\downarrow$ |
| MatMCD (MAS)          | .71 $\pm$ .06 | .74 $\pm$ .02 | .12 $\pm$ .01    | 2.3 $\pm$ 1.3    | .10 $\pm$ .01    | 538.0 $\pm$ 42.5  | .69 $\pm$ .12     | .60 $\pm$ .02 | .66 $\pm$ .06 | .10 $\pm$ .01    | 5.0 $\pm$ 1.0    | .15 $\pm$ .01    | 749.0 $\pm$ 65.3  | 1.05 $\pm$ .15    |
| Base Agent            | .58 $\pm$ .07 | .67 $\pm$ .05 | .20 $\pm$ .05    | 3.0 $\pm$ .8     | .15 $\pm$ .04    | 165.7 $\pm$ 35.9  | 2.00 $\pm$ .54    | .55 $\pm$ .05 | .56 $\pm$ .01 | .13 $\pm$ .04    | 5.5 $\pm$ .5     | .18 $\pm$ .02    | 203.0 $\pm$ 70.5  | 2.42 $\pm$ .26    |
| Evoskill              | .62 $\pm$ .05 | .70 $\pm$ .03 | .17 $\pm$ .03    | 2.5 $\pm$ .5     | .13 $\pm$ .03    | 193.7 $\pm$ 44.2  | 2.03 $\pm$ .63    | .67 $\pm$ .05 | .74 $\pm$ .03 | .10 $\pm$ .02    | 3.5 $\pm$ .5     | .12 $\pm$ .02    | 205.0 $\pm$ 33.8  | 2.48 $\pm$ .28    |
| MAS Compiler          | .62 $\pm$ .07 | .70 $\pm$ .03 | .17 $\pm$ .03    | 2.5 $\pm$ .5     | .13 $\pm$ .03    | 52.3 $\pm$ 8.0    | .82 $\pm$ .08     | .59 $\pm$ .03 | .64 $\pm$ .08 | .12 $\pm$ .03    | 4.7 $\pm$ .6     | .16 $\pm$ .02    | 96.8 $\pm$ 14.7   | 1.08 $\pm$ .02    |
| Tools Only            | .52 $\pm$ .03 | .65 $\pm$ .02 | .24 $\pm$ .03    | 3.7 $\pm$ .5     | .18 $\pm$ .02    | 112.2 $\pm$ 20.2  | .43 $\pm$ .09     | .52 $\pm$ .03 | .54 $\pm$ .05 | .13 $\pm$ .01    | 5.7 $\pm$ .5     | .19 $\pm$ .02    | 123.1 $\pm$ 64.9  | .45 $\pm$ .11     |
| Knowledge Only        | .57 $\pm$ .03 | .67 $\pm$ .02 | .20 $\pm$ .02    | 3.0 $\pm$ .4     | .15 $\pm$ .02    | 22.9 $\pm$ 1.8    | .47 $\pm$ .08     | .61 $\pm$ .05 | .74 $\pm$ .06 | .15 $\pm$ .02    | 4.0 $\pm$ .8     | .13 $\pm$ .03    | 30.2 $\pm$ 2.7    | .49 $\pm$ .09     |
| Pipeline Only         | .57 $\pm$ .02 | .67 $\pm$ .03 | .20 $\pm$ .01    | 3.0 $\pm$ .5     | .15 $\pm$ .01    | 22.6 $\pm$ 1.0    | .44 $\pm$ .07     | .50 $\pm$ .07 | .61 $\pm$ .10 | .19 $\pm$ .02    | 6.0 $\pm$ 1.4    | .20 $\pm$ .05    | 32.0 $\pm$ 5.7    | .42 $\pm$ .08     |
| AdaSkill w/o Opt.     | .52 $\pm$ .03 | .63 $\pm$ .02 | .24 $\pm$ .03    | 3.7 $\pm$ .5     | .18 $\pm$ .02    | 105.7 $\pm$ 13.1  | 1.48 $\pm$ .17    | .58 $\pm$ .07 | .59 $\pm$ .07 | .11 $\pm$ .02    | 5.0 $\pm$ .8     | .17 $\pm$ .03    | 99.3 $\pm$ 19.0   | 1.59 $\pm$ .45    |
| AdaSkill (Ours)       | .56 $\pm$ .08 | .65 $\pm$ .05 | .22 $\pm$ .06    | 3.3 $\pm$ .9     | .17 $\pm$ .05    | 112.0 $\pm$ 1.2   | 1.99 $\pm$ .31    | .65 $\pm$ .07 | .70 $\pm$ .05 | .11 $\pm$ .04    | 4.0 $\pm$ .8     | .13 $\pm$ .03    | 169.3 $\pm$ 51.8  | 2.62 $\pm$ .57    |

| Dataset $\rightarrow$ | Sachs         |               |                  |                  |                  |                    |                   | Child         |               |                  |                  |                  |                    |                   |
|-----------------------|---------------|---------------|------------------|------------------|------------------|--------------------|-------------------|---------------|---------------|------------------|------------------|------------------|--------------------|-------------------|
| Method $\downarrow$   | P $\uparrow$  | F1 $\uparrow$ | FPR $\downarrow$ | SHD $\downarrow$ | NHD $\downarrow$ | Time $\downarrow$  | Cost $\downarrow$ | P $\uparrow$  | F1 $\uparrow$ | FPR $\downarrow$ | SHD $\downarrow$ | NHD $\downarrow$ | Time $\downarrow$  | Cost $\downarrow$ |
| MatMCD (MAS)          | .58 $\pm$ .10 | .56 $\pm$ .11 | .06 $\pm$ .01    | 13.3 $\pm$ 100.3 | .11 $\pm$ .01    | 2747.0 $\pm$ 185.5 | 3.82 $\pm$ .45    | .65 $\pm$ .06 | .65 $\pm$ .05 | .02 $\pm$ .01    | 14.3 $\pm$ 80.3  | .03 $\pm$ .01    | 3774.0 $\pm$ 256.2 | 6.49 $\pm$ .72    |
| Base Agent            | .86 $\pm$ .03 | .87 $\pm$ .05 | .03 $\pm$ .01    | 4.0 $\pm$ 1.0    | .04 $\pm$ .01    | 153.6 $\pm$ 2.9    | 1.78 $\pm$ .53    | .77 $\pm$ .17 | .80 $\pm$ .17 | .02 $\pm$ .01    | 9.3 $\pm$ 7.4    | .03 $\pm$ .02    | 201.8 $\pm$ 51.4   | 2.66 $\pm$ .52    |
| Evoskill              | .86 $\pm$ .08 | .87 $\pm$ .08 | .03 $\pm$ .02    | 3.5 $\pm$ 1.7    | .03 $\pm$ .02    | 154.0 $\pm$ 39.9   | 1.39 $\pm$ .13    | .88 $\pm$ .13 | .90 $\pm$ .10 | .01 $\pm$ .01    | 5.0 $\pm$ 5.0    | .01 $\pm$ .01    | 457.2 $\pm$ 133.2  | 4.83 $\pm$ .72    |
| MAS Compiler          | .75 $\pm$ .11 | .67 $\pm$ .26 | .03 $\pm$ .01    | 7.7 $\pm$ 4.6    | .07 $\pm$ .04    | 198.0 $\pm$ 109.1  | 1.95 $\pm$ 1.11   | .86 $\pm$ .05 | .86 $\pm$ .05 | .01 $\pm$ .01    | 6.7 $\pm$ 3.5    | .02 $\pm$ .01    | 328.9 $\pm$ 131.5  | 2.48 $\pm$ .65    |
| Tools Only            | .86 $\pm$ .08 | .89 $\pm$ .08 | .03 $\pm$ .02    | 3.5 $\pm$ 1.7    | .03 $\pm$ .02    | 92.9 $\pm$ 62.9    | .29 $\pm$ .05     | .90 $\pm$ .05 | .85 $\pm$ .03 | .01 $\pm$ .01    | 5.7 $\pm$ .9     | .01 $\pm$ .01    | 115.6 $\pm$ 23.2   | .54 $\pm$ .15     |
| Knowledge Only        | .88 $\pm$ .05 | .89 $\pm$ .05 | .02 $\pm$ .01    | 3.3 $\pm$ 1.3    | .03 $\pm$ .01    | 54.9 $\pm$ 9.4     | .41 $\pm$ .07     | .81 $\pm$ .08 | .79 $\pm$ .07 | .01 $\pm$ .01    | 9.3 $\pm$ 3.3    | .03 $\pm$ .01    | 161.4 $\pm$ 103.2  | .54 $\pm$ .11     |
| Pipeline Only         | .94 $\pm$ .05 | .95 $\pm$ .05 | .01 $\pm$ .01    | 1.7 $\pm$ 1.7    | .02 $\pm$ .02    | 75.9 $\pm$ 65.1    | .45 $\pm$ .08     | .79 $\pm$ .05 | .81 $\pm$ .04 | .02 $\pm$ .01    | 9.0 $\pm$ 2.2    | .02 $\pm$ .01    | 55.0 $\pm$ 18.2    | .77 $\pm$ .14     |
| AdaSkill w/o Opt.     | .93 $\pm$ .03 | .95 $\pm$ .03 | .01 $\pm$ .01    | 1.3 $\pm$ .5     | .01 $\pm$ .01    | 237.9 $\pm$ 92.0   | 2.51 $\pm$ .82    | .88 $\pm$ .03 | .84 $\pm$ .05 | .01 $\pm$ .01    | 5.7 $\pm$ 1.2    | .02 $\pm$ .01    | 158.8 $\pm$ 38.4   | 1.96 $\pm$ .21    |
| AdaSkill (Ours)       | .91 $\pm$ .07 | .92 $\pm$ .06 | .02 $\pm$ .01    | 2.3 $\pm$ 1.7    | .02 $\pm$ .02    | 301.8 $\pm$ 212.5  | 2.82 $\pm$ 1.11   | .96 $\pm$ .06 | .95 $\pm$ .07 | .00 $\pm$ .00    | 1.7 $\pm$ 2.4    | .00 $\pm$ .01    | 170.6 $\pm$ 12.7   | 2.55 $\pm$ .62    |





**Figure 13: Stage 2 iterator trajectories validate  $F$ -dependent convergence behavior.** Low- $F$  CE-MSA (a) gains quickly then oscillates, illustrating the knife-edge landscape risk. Mid-to-high- $F$  tasks—CD (b), T2SQL (c), and FE (d)—improve steadily and plateau cleanly, confirming that safe monotonic refinement is achievable precisely where Stage 1 leaves the most headroom. Solid = val; dashed = train; green line = selected best version.

## E Stage 2: Skill Iterator Details

Fig. 13 tracks Stage 2 optimization across all four domains; the trajectories align with Theorem 3.1 and the main-text summary in Section 4.6.

On **CE-MSA** ( $F \approx 0$ ), the iterator quickly reaches 100% validation accuracy by iteration 2 via tool fixes and knowledge alignment. However, iterations 3–4 exhibit classic oscillation—fixing one failure case breaks another. The knife-edge landscape turns optimization into a zero-sum game, validating why Stage 2 is disabled for low- $F$  metrics.

On **CD** ( $F = 0.24\text{--}0.77$ ) and **T2SQL** ( $F = 0.50$ ), the forgiving scoring surface allows the iterator to surgically inject tool fixes and missing knowledge without displacing correct behaviors. CD gains +14pp on DWDCClimate ( $F1=0.701$ ) and +15pp on Child ( $F1=0.952$ ); T2SQL val EX improves steadily, plateauing at 86.7% from v3 on BIRD-147. Both confirm smooth, monotonic hill-climbing in mid-freedom regimes.

On **FE** ( $F \approx 0.59\text{--}0.97$ ), val AUC on Taobao plateaus after iteration 2 with only marginal gains. The flat landscape means structural tweaks cannot meaningfully improve the score, as the base agent’s intrinsic capability ceiling has already been reached.

### E.1 Iteration-by-Iteration Results

Table 10 shows the per-iteration metrics and key changes during Stage 2 refinement on Causal Estimation (CE-MSA), corresponding to panel (a) of Figure 13.

**Table 10:** Skill iterator: iteration-by-iteration improvements on Causal Estimation.

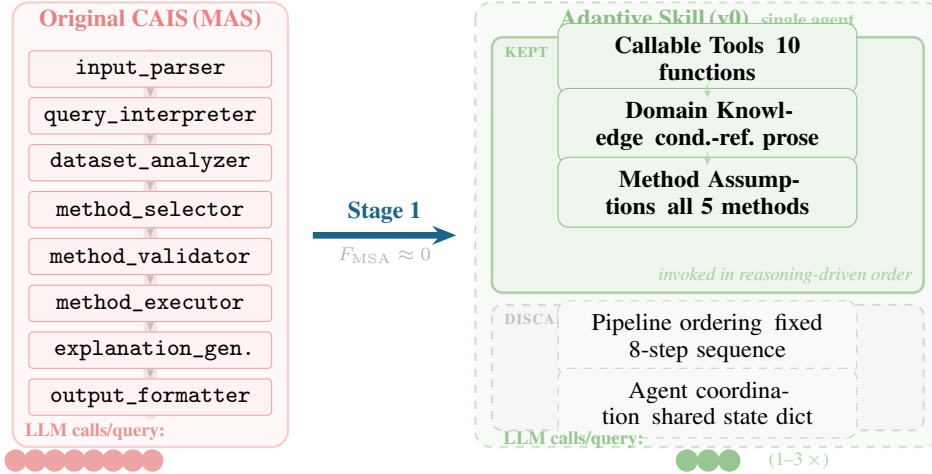
| Iter | Skill            | Tr. MSA     | Val MSA     | Tr. MRE      | Val MRE      | Key Change                           |
|------|------------------|-------------|-------------|--------------|--------------|--------------------------------------|
| 0    | v0 (base)        | 100%        | 60%         | 51.4%        | 65.7%        | Baseline                             |
| 1    | v1               | 100%        | 80%         | 13.8%        | 44.9%        | Tool fix: DiD/RDD sign; IV rules     |
| 2    | <b>v2 (best)</b> | <b>100%</b> | <b>100%</b> | <b>30.6%</b> | <b>21.9%</b> | Thinking: RCT vs. observational      |
| 3    | v3               | 100%        | 80%         | 31.8%        | 44.9%        | IV fix broke covariate rules         |
| 4    | v4               | 100%        | 80%         | 13.2%        | 33.5%        | treatment_above; Broockman regressed |
| 5    | v5               | 100%        | 80%         | 31.8%        | 31.8%        | Oscillation: fix one → break another |

## F Skill Evolution Case Study: Causal Estimation

We trace the complete transformation from CAIS (the original MAS) to the final optimized skill, illustrating the design decisions in Stage 1 and the concrete rule-level fixes in Stage 2. This case study corresponds to the CE-MSA results in Table 2.

### Stage 1: Adaptive Distillation — Architecture Transformation

Figure 14 contrasts the original CAIS 8-step pipeline with the resulting single-agent skill.



**Figure 14: Stage 1 architecture transformation for CE** ( $F_{\text{MSA}} \approx 0$ ). **Left:** the original CAIS MAS has 8 LLM-backed agents on a shared state dict (8 red dots = 8 sequential LLM calls/query). **Right:** the adaptive skill retains tools and knowledge as freely-invocable layered modules and discards pipeline ordering and agent coordination. LLM calls reduce to 1–3× while CE-MSA gains +28 pp.

**What changes in the representation of knowledge.** Beyond architecture, the *form* of domain knowledge changes. The original `method_selector` encodes decision logic as hard-coded Python:

Original MAS: `rule_based_select_method` (excerpt)

```
if is_rct:
    if instrument_var and instrument_var != treatment:
        return {"selected_method": "instrumental_variable", ...}
    if covariates:
        return {"selected_method": "linear_regression", ...}
    else:
        return {"selected_method": "diff_in_means", ...}
```

The adaptive skill converts the same logic to conditional-reference prose in `SKILL.md`:

v0 `SKILL.md`: same knowledge as conditional reference

```
### Step 1 - RCT check (do this first)
If description contains: randomized / RCT / randomly assigned -> is_rct = True
- When is_rct = True: matching, PSM, and IPW are FORBIDDEN.
  OLS with covariates is the correct choice.
```

```

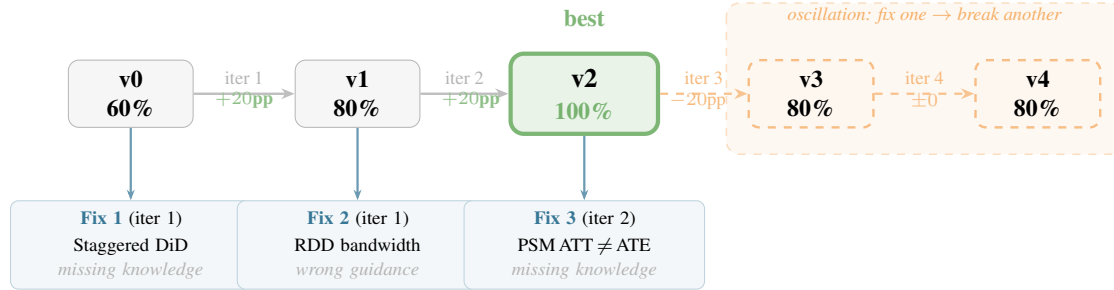
Random assignment already balances covariates --
applying matching on top produces an incorrect selected_method.
- If an instrument for non-compliance exists -> use IV (encouragement design).

```

The agent now reasons about the rule in context rather than executing it mechanically. This is the key shift: *the same factual content, zero pipeline structure.*

## Stage 2: Iterator Trace — Three Targeted Fixes

Figure 15 shows the Val MSA trajectory and the specific SKILL.md rules added or changed at each iteration.



**Figure 15: Stage 2 iterator trajectory for CE-MSA.** Version boxes show Val MSA; v2 (green, bold border) is the globally best version. Iteration 1 applies two concurrent fixes (+20 pp, 60% → 80%); iteration 2 applies one further fix (+20 pp, 80% → 100%). Iterations 3–4 enter oscillation (orange shaded region): each patch repairs one rule but inadvertently breaks another, yielding no net gain. The iterator identifies v2 and terminates.

**Fix 1 (Iteration 1): Staggered DiD.** *Failing case:* panel dataset, 57 unique treatment-start years (first\_sanction). The agent set treatment\_period=1982 (the mean), so units treated before 1982 received \_post = 0 for their actual treatment periods — a mis-specified TWFE.

**Before (v0)** — no guidance on staggered adoption:

```

### DiD
- treatment_group_col: binary column (1 = treated group, 0 = control)
- treatment_period: the single period when treatment started

```

**After (v1)** — explicit staggered TWFE path added:

```

### CRITICAL: Two types of DiD -- choose correctly
Type B -- Staggered DiD: dataset has BOTH an ever-treated column
(ever_sanctioned) AND a period-specific column (treat).
-> Do NOT use estimate_did. Use estimate_ols with treat as treatment
+ unit FE + time FE. Staggered signal: first_treatment_year
with many unique values.

```

**Fix 2 (Iteration 1): RDD bandwidth.** *Failing case:* Head Start RDD,  $n=23,599$ . Agent used 10%-of-range bandwidth → estimate =  $-0.031$ ; reference =  $-0.079$ ; MRE = 61%.

**Before (v0):**

```

### RDD
- Bandwidth choice: try default (10% of range), then check n_in_bandwidth >= 20

```

**After (v1):**

```

### RDD -- bandwidth (critical for accuracy)
ALWAYS call estimate_rdd with bandwidth=None.
The tool uses rdrobust MSE-optimal IK bandwidth automatically
(same algorithm as R reference). Do NOT specify a narrow bandwidth.

```

**Fix 3 (Iteration 2): PSM ATT  $\neq$  ATE.** *Failing case:* social media campaign, max\_smd=1.01, 50/50 split. Agent ran PSM (correct method choice), but PSM ATT= 2.23 vs. ATE reference= 1.58; MRE= 41.3%.

**Before (v1)** — warning exists, no action:

```
### Matching / PSM
- Estimates ATT (effect on treated), NOT ATE -- this is critical
```

**After (v2)** — mandatory AIPW step added:

```
### Step 3b -- Matching workflow (mandatory)
When matching selected (max_smd > 0.3):
  1. Run check_overlap (verify common support)
  2. Call estimate_aipw for final ATE <-- NEW
    AIPW = doubly-robust augmented IPW; estimates population ATE.
    PSM estimates ATT only -- using PSM as ATE gives ~40% MRE
    when treated/control populations differ.
    Do NOT report estimate_psm output as the final ate value.
```

The `estimate_aipw()` function was added to `estimators.py` with bootstrap SE.

**Summary.** Table 11 compares the original MAS, v0, and v2. The v0→v2 changes are small (3 targeted edits, ~20 new lines) yet produce a +40pp Val MSA gain and add a new callable tool—illustrating the Analyzer’s surgical, trace-driven design.

**Table 11:** CAIS MAS → Skill evolution across Stage 1 and Stage 2.

| Dimension         | Original MAS           | v0 (Stage 1)          | v2 (Stage 2)               |
|-------------------|------------------------|-----------------------|----------------------------|
| Workflow          | 8-step forced pipeline | Free tool invocation  | Free tool invocation       |
| Method selection  | Hard-coded rule tree   | Conditional-ref prose | Conditional-ref prose      |
| DiD support       | Canonical only         | Canonical + hint      | Canonical + staggered TWFE |
| RDD bandwidth     | 10% range fallback     | 10% range fallback    | rdrobust IK-optimal        |
| Matching estimand | PSM (ATT)              | PSM (ATT)             | AIPW (population ATE)      |
| Val MSA           | —                      | 60%                   | <b>100%</b>                |
| Cost / query      | \$0.086                | \$0.169               | \$0.194                    |